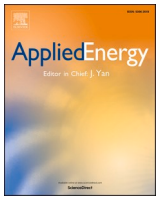
Applied Energy 399 (2025) 126442



Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy





# Comprehensive evaluation of all-element flexibility resources in data centers: considering synergistic benefits of computing, electricity, and heat

Yongzhen Wang [a,b,c], Jiayu Lin [a], Yibo Han [a], Kai Han [a,b,*], Juntao Han [a], Te Han [c,d], Yiming Wei [c,d]

[a] School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China
[b] Innovation Center in Chongqing, Beijing Institute of Technology, Chongqing 401120, China
[c] Beijing Laboratory for System Engineering of Carbon Neutrality, Beijing 100081, China
[d] School of Management, Beijing Institute of Technology, Beijing 100081, China

## HIGHLIGHTS

- This paper established a collaborative operation model for flexibility resources.
- An optimization method for all-element flexibility resources is developed.
- The impact of each flexibility resource on multiple metrics is revealed.

## ARTICLE INFO

*Keywords:*
Data center
Flexibility resources
Computing-heat coupling
Capacity planning and operation scheduling

## ABSTRACT

The exploration of internal resource flexibility in data centers is a significant trend for energy conservation, emission reduction, and sustainable development. To systematically quantify the coupling mechanisms and potential of flexibility in the computing, electricity, and heat side, this paper develops a multi-element flexibility resource evaluation method integrated capacity planning and operation scheduling for data centers. It reveals the impact of each flexibility resource on the data center's energy consumption, carbon emissions, economic costs, Power Usage Effectiveness (PUE), computing efficiency, and other integrated performance metrics. An optimization and scheduling framework for all-element flexibility resources is developed based on a collaborative operation model for flexibility resources established in this paper. The results show that the flexibility is able to reduce carbon emissions by 38.5 %, energy consumption by 22.8 %, and economic costs by 21.6 %, while significantly improving PUE and computing efficiency. In particular, the utilization of electricity flexibility notably reduces carbon emissions by 21.5 %, while the coordinated regulation of computing flexibility and heat flexibility has a significant effect on reducing energy consumption (by 11.1 % and 13.1 %, respectively). Furthermore, the impact of factors like time-of-use tariff, workload status, and rack occupancy rate on flexibility resource utilization is analyzed through sensitivity analysis, revealing how these factors influence the optimization of the data center's performance metrics.

## 1. Introduction

With the rapid development of "Internet Plus", the scale of data centers, which are key computing infrastructure supporting the digital economy, has significantly increased in recent years. The number of computing racks has also surged, leading to a rise in the overall operational costs and energy consumption of data centers(DCs). It is estimated that by 2030, the total energy consumption of global data centers will reach around 848 TWh [ [1]]. At the same time, the electricity required for the operation of DCs is accompanied by significant carbon dioxide emissions, which will have a considerable impact on global sustainable development. Unlike conventional power-side loads, DC electricity loads exhibit considerable flexibility [ [2]]. By deeply exploring the flexibility of DC power systems, it is possible to effectively reduce the

* Corresponding author at: School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China.
*E-mail addresses:* wyz80hou@bit.edu.cn (Y. Wang), autosim@bit.edu.cn (K. Han), hanjt0506@bit.edu.cn (J. Han), hante@bit.edu.cn (T. Han), wei@bit.edu.cn (Y. Wei).

https://doi.org/10.1016/j.apenergy.2025.126442
Received 28 January 2025; Received in revised form 2 July 2025; Accepted 7 July 2025
Available online 21 July 2025
0306-2619/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

energy consumption, carbon emissions, and economic costs of DCs, transforming the growing electricity load of DCs into a flexible resource for the grid rather than a burden.

The International Energy Agency (IEA) defines power system flexibility as [ [3]]: the ability of a power system to rapidly respond to significant fluctuations in supply and demand within its boundary constraints, quickly reacting to both predictable and unpredictable changes and events, decreasing supply when demand decreases, and increasing supply when demand rises. Based on this definition, resources within a DC that can provide or enhance such capabilities are termed "flexibility resources". In conventional DCs, flexibility resources can primarily be divided into three categories: computing-side flexibility resources, mainly including the spatiotemporal transferability of workload, and the adjustability of server energy consumption per computational power; electricity-side flexibility resources, mainly including backup power, energy storage devices, and their interactions with the grid market; and heat-side flexibility resources, mainly including the adjustability of inlet temperature of the racks and its impact on the cooling system, building thermal inertia, and virtual energy storage characteristics.

Currently, several studies have focused on the application of flexibility resources in DCs, as shown in Table 1. In research on the computing side, Cao et al. [4] proposed a data-driven flexibility evaluation scheme for batch workload in DCs, demonstrating that utilizing the temporal transfer characteristics of batch workload can effectively regulate the electricity load of DCs. Jian et al. [5] and Wu et al. [6] comprehensively considered the temporal and spatial transfer characteristics of the workload, studying temporal transfer within a single DC and spatial transfer scheduling strategies and flexibility between distributed DCs. Chen et al. [7] constructed a set of two-stage spatiotemporal scheduling algorithm to transfer the arithmetic load of off-site DCs, which effectively improves the renewable energy consumption ability of DCs and reduces the carbon emission and cost of DCs. Zhou et al. [8,9] considered the price differences of electricity across different areas and proposed a multi-resource flexibility scheduling strategy for geographically distributed DCs, coordinating both electricity-side flexibility resources and cooling system energy consumption. Dvorkin [10] developed an immediate intelligent scheduling algorithm for workload of servers, considering the large and sensitive characteristics of workload. Most studies focus on the flexibility of spatial transfer between multiple DCs, while some scholars have also studied the flexibility of load spatial transfer between servers within a DC. Jin et al. [11] summarized commonly used server energy consumption models in current research, with many scholars attempting to fit server energy consumption using aggregation models [12,13] and Baseline power + Active power (BA) models [14,15]. For example, Fan et al. [16] proposed that the overall power consumption of a server lies between idle and peak power consumption, with a linear relationship to the load rate. However, Zhang et al. [17] found that the relationship between server energy consumption and load rate is more consistent with a cubic curve based on SPEC power benchmark test data [18]. Lin et al. [19] further designed a heuristic task scheduling algorithm based on this nonlinear relationship, proving that the algorithm can effectively reduce energy consumption in server clusters, although it does not consider the temporal transferability of workload.

In addition to server energy consumption, the cooling system in DCs accounts for approximately 38 % of total energy consumption, forming the primary source of energy consumption alongside IT equipment [ [20]]. Moreover, the cooling system not only serves as a heat carrier for IT equipment but also plays a critical role in affecting the energy consumption of IT equipment and the overall DC. As a result, many studies have begun to focus on heat flexibility and the synergistic relationship between electricity and heat flexibility in enhancing DC energy efficiency. For instance, Cui et al. [21] proposed a capacity configuration scheme that considers factors such as the time transferability of DC load, air thermal inertia, and waste heat recovery. Ji et al. [22] optimized the coordination between the cooling system and IT equipment's energy consumption with PUE as the optimization objective, aiming to reduce the overall energy consumption of the DC. Damme et al. [23] coordinated and optimized the cooling system output and server computational load distribution using thermal-aware optimization technology. However, most of these studies focus on detailed modeling of a single flexibility resource and lack in-depth discussion of the coupling relationships between different flexibility resources. Wang et al. [24] developed a detailed cooling system model based on thermal inertia and environmental temperature, considering the flexibility of both IT equipment and cooling systems to reduce operating costs and improve energy efficiency. In addition to virtual energy storage, Li et al. [25,26] further considered the relationship between waste heat recovery in DCs and server energy consumption and voltage frequency, expanding the boundary of heat flexibility while also exploring in depth the coupling relationship between computing flexibility and electricity flexibility. However, they did not fully consider the impact of temperature regulation on server performance and energy consumption. Ham et al. [27] studied the dynamics of central processing unit (CPU) power consumption and leakage power, exploring the mechanism by which

**Table 1**
Studies focused on the application of flexibility resources in DCs (The Partial).

| Literature | Computing flexibility | | Heat flexibility | | Electricity flexibility | Computing-heat coupling |
|---|---|---|---|---|---|---|
| | Temporal | Spatial | Thermal inertia | Cooling system | | |
| This Paper | √ | Between servers | √ | √ | √ | √ |
| Cao et al. [4] | √ | × | × | × | √ | × |
| Jian et al. [5] | √ | Between DCs | × | × | √ | × |
| Wu et al. [6] | √ | Between DCs | × | × | × | × |
| Chen et al. [7] | √ | Between DCs | × | × | × | × |
| Zhou et al. [8] | √ | Between DCs | √ | √ | √ | × |
| Zhou et al. [9] | √ | Between DCs | √ | √ | √ | × |
| Dvorkin [10] | √ | Between DCs | × | × | × | × |
| Lin et al. [19] | × | Between servers | × | × | × | × |
| Cui et al. [21] | √ | × | √ | √ | √ | × |
| Ji et al. [22] | √ | Between servers | × | √ | √ | × |
| Damme et al. [23] | × | × | × | √ | × | × |
| Wang et al. [24] | √ | × | √ | √ | × | × |
| Li et al [25] | √ | × | √ | √ | √ | × |
| Li et al [26] | √ | × | √ | √ | √ | × |
| Kamran et al. [32] | √ | Between DCs | × | × | √ | × |

Note: "×" indicates that the paper does not consider this type of flexibility, and "√" indicates the opposite; "Between DCs" and "Between servers" indicates whether the paper considers the spatial transfer potential of workload between DCs or servers.

temperature affects CPU power consumption. Zhu et al. [28] further proposed a model in which both temperature and load rate jointly influence CPU energy consumption. Pagani et al. [29] suggested that when the core temperature of a server rises, internal components such as processors automatically reduce their operating frequency to reduce heat generation, thereby reducing computing performance. Guo et al. [30] further studied the relationship between server computing performance, core temperature, and environmental temperature. These studies tend to focus on the one-way influence of heat flexibility resources on computing flexibility resources or vice versa, without organically integrating computing-side flexibility resources and heat-side flexibility resources, and considering the coupling mechanisms between these two types of flexibility resources.

To address these issues, the main contributions of this paper are as follows:

1. This paper provides a detailed description of the coupling mechanisms between computing-side flexibility resources, heat-side flexibility resources, and electricity-side flexibility resources within an independent DC.
2. A capacity planning and operational scheduling strategy is proposed based on all-element flexibility, aiming to improve the energy efficiency of DCs, reduce carbon emissions, and lower economic costs. The impact and mechanisms of all-element flexibility on indicators such as PUE and computing efficiency are also explored.
3. Sensitivity analysis is conducted for three major factors that significantly affect flexibility resource utilization: time-of-use tariffs, computing workload types, and DC rack occupancy rate. The analysis examines the impact on the electricity, computing, and heat flexibility, further studying and analyzing the utilization mechanisms of the three types of flexibility resources in DCs, providing a theoretical basis for optimized scheduling in DCs.

## 2. Framework description and modeling

The overall framework of the research object is shown in Fig. 1. For conventional DCs, their primary function is to receive and process workload from users in real-time. The required electricity is supplied by the grid, and heat is dissipated through the cooling system to maintain the temperature of the DC within a specific range, aiming to keep it as constant as possible. However, in reality, DCs contain a large amount of flexible resources that can be utilized. These resources can be divided into three main categories: computing-side flexibility resources, electricity-side flexibility resources, and heat-side flexibility resources.

Electricity flexibility is represented by the interaction between backup power sources (this paper mainly considers gas turbines as backup power) and energy storage devices with the peak and off-peak electricity prices from the grid. Computing flexibility is demonstrated by the temporal transferability of batch workload, the spatial transferability of interactive workload between servers, and the adjustability of energy consumption per computational power in relation to the load rate. Heat flexibility mainly manifests in the coefficient of performance (COP) of the cooling system (this study focuses on air-cooled DCs) as it changes with the air supply temperature, as well as the virtual energy storage in the DC building. Meanwhile, the electricity-side energy supply equipments provide electrical energy for the computing equipment on the computing side and the cooling system on the heat side; the computing-side load directly affects the cold load of the cooling system on the heat side, while the cooling system on the heat side indirectly affects the computing-side load by regulating the temperature. Modeling is required to more accurately and quantitatively describe the synergistic relationship between the three-side flexibility resources.

### 2.1. Electricity flexibility

On the electricity side, this paper mainly considers the flexible scheduling capabilities of the backup power and energy storage devices in DCs. The backup power source considered in this study is a gas turbine, which provides electricity to the system by consuming natural gas and, together with the large number of energy storage devices equipped in the DC (currently, most DCs use electrochemical batteries such as lead-acid batteries or lithium batteries), participates in the flexible

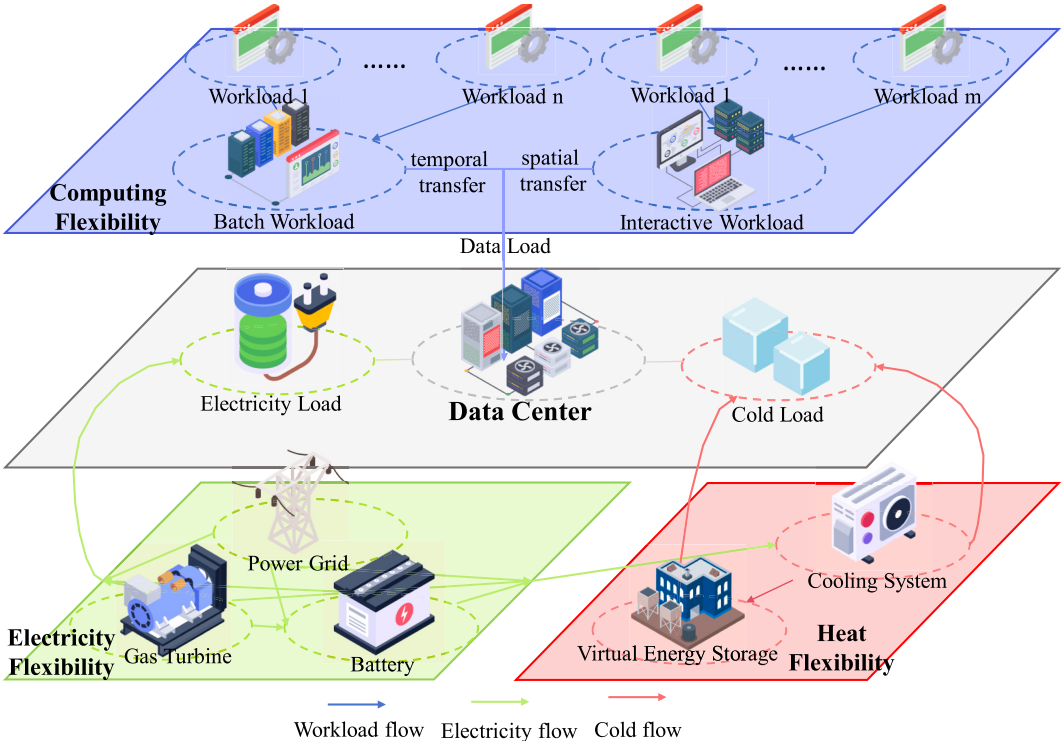



**Fig. 1.** The overall framework of the flexibility resources in a DC.

scheduling of the power system. In the operation of traditional DCs, gas turbines and power storage devices are regarded as backup power sources, which are used when, and only when, there is an emergency such as a power outage in the DC, and only rely on the grid for power supply during normal operation. However, DCs will reserve a large amount of backup power for the sake of DC operation reliability. This is actually a waste of resources, so this paper regards this part of unused "backup power" in the DC as a resource that can be flexibly utilized without affecting the reliable operation of the DC.

For a DC, the energy supply method can be switched flexibly based on its own needs, and the electricity load is jointly met by the grid, gas turbines, and energy storage devices:

$$P_t^{\mathrm{DC}} = P_t^{\mathrm{grid}} + P_t^{\mathrm{gt}} + P_t^{\mathrm{ba,out}} - P_t^{\mathrm{ba,in}} \tag{1}$$

where: $P_t^{\mathrm{DC}}$ is the total electricity demand of the DC at time $t$; $P_t^{\mathrm{grid}}$ is the power output from the grid; $P_t^{\mathrm{gt}}$ is the power output from the gas turbine; $P_t^{\mathrm{ba,out}}$ and $P_t^{\mathrm{ba,in}}$ are the output and input of the electrochemical batteries, respectively.

The operational constraints of the gas turbine are as follows [31]:

$$\begin{cases} P_t^{\mathrm{gt}} = \eta^{\mathrm{gt}} \lambda^{\mathrm{gas}} f_t^{\mathrm{gt}} \\ \eta^{\mathrm{gt}} = g_1 x_{\mathrm{gt}}^3 + g_2 x_{\mathrm{gt}}^2 + g_3 x_{\mathrm{gt}} + g_4 \\ 0 \le P_t^{\mathrm{gt}} \le P^{\mathrm{c,gt}} \\ \Delta P_{\min}^{\mathrm{gt}} \le P_t^{\mathrm{gt}} - P_{t-1}^{\mathrm{gt}} \le \Delta P_{\max}^{\mathrm{gt}} \\ P_{\min}^{\mathrm{c,gt}} \le P^{\mathrm{c,gt}} \le P_{\max}^{\mathrm{c,gt}} \end{cases} \tag{2}$$

where: $\eta^{\mathrm{gt}}$ is the efficiency of the gas turbine, related to the load rate $x_{\mathrm{gt}}$ of gas turbine, fitted with the real gas turbine operational data; $g_1 \sim g_4$ are the fitting coefficients used to characterize the relationship between the power efficiency and load rate of the gas turbine; $\lambda^{\mathrm{gas}}$ is the calorific value of natural gas; $f_t^{\mathrm{gt}}$ is the natural gas consumption rate; $P^{\mathrm{c,gt}}$ is the gas turbine's power capacity; $\Delta P_{\min}^{\mathrm{gt}}$ and $\Delta P_{\max}^{\mathrm{gt}}$ are the upper and lower limits of GT climb power, respectively;$P_{\min}^{\mathrm{c,gt}}$ and $P_{\max}^{\mathrm{c,gt}}$ are the lower and upper limits of the installed capacity of the gas turbine, the lower limit of the installed capacity of the gas turbine is set to be higher than the peak power of the DC for the season of reliability.

The operational constraints of the electrochemical battery are as follows [29]:

$$\begin{cases} S_{t+1}^{\mathrm{ba}} = S_t^{\mathrm{ba}} + \eta^{\mathrm{ba,in}} P_t^{\mathrm{ba,in}} - P_t^{\mathrm{ba,out}} / \eta^{\mathrm{ba,out}} \\ S_{\min}^{\mathrm{ba}} \le S_t^{\mathrm{ba}} \le S_{\max}^{\mathrm{ba}} \\ 0 \le P_t^{\mathrm{ba,in}} \le P_{\max}^{\mathrm{ba,in}} \\ 0 \le P_t^{\mathrm{ba,out}} \le P_{\max}^{\mathrm{ba,out}} \\ P_t^{\mathrm{ba,in}} P_t^{\mathrm{ba,out}} = 0 \end{cases} \tag{3}$$

where: $S_t^{\mathrm{ba}}$ is the energy storage capacity of the electrochemical battery at time $t$;$\eta^{\mathrm{ba,in}}$ and $\eta^{\mathrm{ba,out}}$ are the charge and discharge efficiencies of the electrochemical battery, respectively; $S_{\min}^{\mathrm{ba}}$ and $S_{\max}^{\mathrm{ba}}$ are the minimum and maximum energy storage capacity of the storage system, determined by the safe operating domain of the DC, the minimum energy storage capacity of the storage system should always be able to meet the electricity load of DC for at least 15 min; $P_{\max}^{\mathrm{ba,in}}$ and $P_{\max}^{\mathrm{ba,out}}$ are the maximum charge and discharge power of the electrochemical battery.

Under the premise of not affecting the reliable operation of DCs, there is a complementary relationship between gas turbines and power grids, especially considering the time-varying characteristics of grid electricity prices, gas turbines that can stably provide energy and be flexibly dispatched can well realize the function of replacing grid power to supplement power supply at times of high electricity prices, so as to enhance the overall economy of DCs and low-carbon; the flexibility of power storage device is mainly reflected in the temporal flexibility, which is achieved by storing the power supplied by the grid at low electricity prices and supplying power at times of high electricity prices, thus regulating the time of the DC's energy demand, realizing the "peak shaving and filling in the valley".

### 2.2. Computing flexibility

The computing flexibility in DCs is mainly reflected in the adjustability of server energy consumption. On one hand, the spatiotemporal transferability of workload can alter the spatiotemporal distribution of energy consumption. On the other hand, the variability in the energy consumption per computational power (i.e., the energy required by a server for processing the same workload) allows for the possibility of adjusting server energy consumption. In this paper, we use the SPECpower2008 software to calculate the energy consumption and comprehensive energy consumption per computational power (CECPCP) of the Huawei 1288H V5 servers used in the designed IT room under different load rates, with the results shown in Fig. 2.

The results show that within the load-bearing range of the server, there is an optimal load rate at which the CECPCP is minimized. However, the server energy consumption tested by the software is based solely on measurements taken under the same ambient temperature and does not account for the impact of core temperature changes on server computing performance. As the server load rate increases, the heat dissipation also increases. During this process, both the load rate and the core temperature jointly influence the total power consumption and heat dissipation of the server, with the load rate being adjustable through workload transfer.

To adjust the load rates of each server, it is first necessary to classify and schedule the workload of the DC. The workload in DCs are mainly divided into batch workload and interactive workload. Interactive workload such as network services, online games, etc., its response latency is usually within 1 s, the response to the real-time requirements of the higher, but for the lower requirements of computing resources, through the front-end proxies (front-end proxies) for aggregation, and then through the network between different servers to transfer the distribution of "space" on the transfer of distribution; Batch workload such as scientific computing, economic analysis, image processing, etc., its response time is usually hours, and even up to several days. With a high response delay tolerance, but higher requirements for computing resources, can be delayed to be processed at any time before the deadline for the completion of the temporal flexibility.

#### 2.2.1. Batch workload transfer

If we number and sort all the batch workload received by the DC in one day, the $j$-th batch workload received by the DC can be represented as:

$$L_{j,24\times 1}^{\mathrm{B}} = \left[ l_{j,1}^{\mathrm{B}}, l_{j,2}^{\mathrm{B}} \cdots, l_{j,24}^{\mathrm{B}} \right] \tag{4}$$

$$l_j^{\mathrm{B}} = \sum_k l_{j,k}^{\mathrm{B}} \tag{5}$$

where: $l_{j,t}^{\mathrm{B}}$ represents the $j$-th computational power required by the batch workload at time t without workload transfer; $l_j^{\mathrm{B}}$ represent the total computing power demand of the $j$-th batch workload received by the DC.

For the temporal transfer characteristics and stable computational demands of batch workload, once a batch workload arrives at the DC, it needs to be assigned to a server and undergo delayed processing for a certain period based on its maximum allowable delay time $D_j$.

For the computing power $F_{t,n}$ of the $n$-th server assigned to that batch workload at time $t$, if $l_j^{\mathrm{B}} / F_{t,n} + D_j$ is small than 24, the number of hours in one day, then $l_{j,t}^{\mathrm{B}}$ can be expressed as:

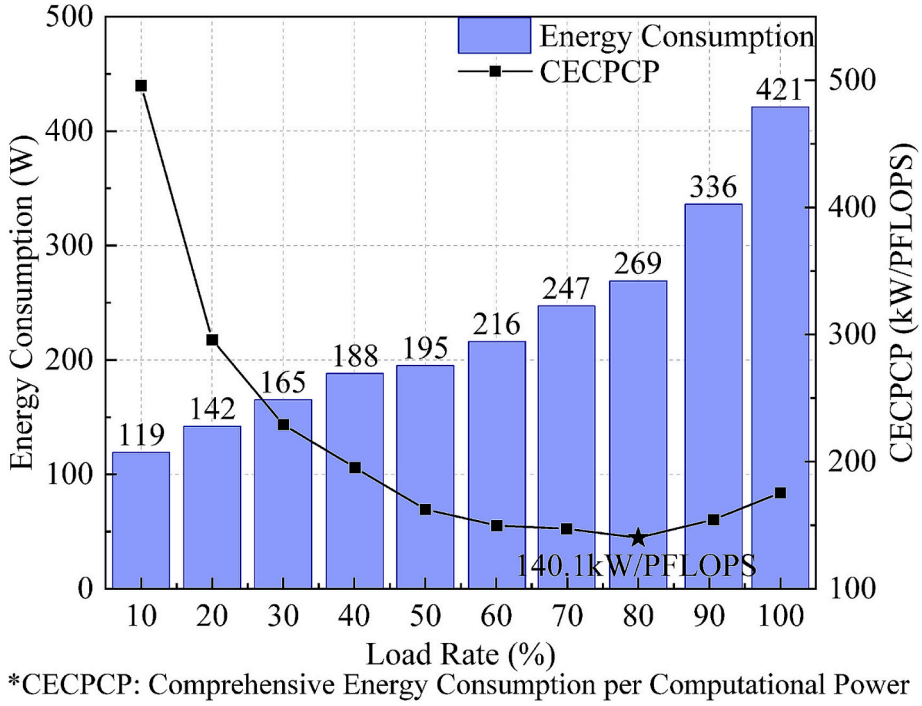



**Fig. 2.** The energy consumption and CECPCP of the Huawei 1288H V5 servers [18].

$$l_{j,t}^{\mathrm{B}}=\begin{cases}0 & if\left(k<D_j\right)\cup t>\left(D_j+l_j^{\mathrm{B}}/F_{t,n}\right)\\ F_{t,n} & if\left(D_j\le t\le\left(l_j^{\mathrm{B}}/F_{t,n}+D_j\right)\right)\cap\left(t+1<l_j^{\mathrm{B}}/F_{t,n}+D_j\right)\\ l_j^{\mathrm{B}}-\left(t-D_j\right)F_{t,n} & if\left(t\le l_j^{\mathrm{B}}/F_{t,n}+D_j\right)\cap\left(t+1\ge l_j^{\mathrm{B}}/F_{t,n}+D_j\right)\end{cases} \tag{6}$$

Otherwise, the $l_{j,t}^{\mathrm{B}}$ can be expressed as:

$$l_{j,t}^{\mathrm{B}}=\begin{cases}0 & if\ l_j^{\mathrm{B}}/F_{t,n}+D_j-24<k<D_j\\ F_{t,n} & if\left(\left(t\ge D_j\right)\cup\left(t\le L_{b,j}/F_{t,n}+D_j-24\right)\right)\cap\left(t+1<l_j^{\mathrm{B}}/F_{t,n}\right)\\ l_j^{\mathrm{B}}-\left(t-D_j\right)F_{t,n} & if\left(t\le l_j^{\mathrm{B}}/F_{t,n}+D_j-24\right)\cap\left(t+1\ge L_{b,j}/F_{t,n}+D_j-24\right)\end{cases} \tag{7}$$

These functions indicate that after the batch workload enters a certain server for computation and no load transfer is performed, the task will occupy the amount of computation power of that server. Since the workload is distributed within a typical day, and assuming that the number and type of IT loads remain unchanged on a daily basis, the computation will continue from the start of the day when the workload cannot be completed in one day, indicating that the workload of the day is carried over to 0:00 h of the next day or that the workload of the previous day is carried over to 0:00 h of the current day or to 0:00 h of the previous day. Workload carries over to time 0 of the current day. To simplify the logical process, the default into the batch load does not need to occupy the rack 100 % load more than 24 h, if more than it is necessary to transfer the cycle to be extended, its distribution method is the same.

After the transfer, the batch workload on each server can be represented as:

$$L^{B}=\begin{bmatrix} x_1^{\mathrm{B,server}}\sum_{j}^{J}\sum_{d}^{24}x_{j,1,d}^{\mathrm{B,time}}l_{j,1}^{\mathrm{B}} & \cdots & x_1^{\mathrm{B,server}}\sum_{j}^{J}\sum_{d}^{D_j}x_{j,24,d}^{\mathrm{B,time}}l_{j,24}^{\mathrm{B}}\\ \vdots & \ddots & \vdots\\ x_{\mathrm{N}}^{\mathrm{B,server}}\sum_{j}^{J}\sum_{d}^{D_j}x_{j,1,d}^{\mathrm{B,time}}l_{j,1}^{\mathrm{B}} & \cdots & x_{N}^{\mathrm{B,server}}\sum_{j}^{J}\sum_{d}^{D_j}x_{j,24,d}^{\mathrm{B,time}}l_{j,24}^{\mathrm{B}}\end{bmatrix} \tag{8}$$

where: $J$ represents the total amount of batch workload; N represents the total number of servers; $x_n^{\mathrm{B,server}}$ is the batch workload spatial transfer coefficient, which is a binary variable, equal to 1 when the workload is assigned to the server numbered $n$; $x_{j,t,d}^{\mathrm{B,time}}$ is the temporal transfer coefficient which is a constant between 0 and 1, indicating the proportion of the $j$-th computing workload required at time $t$ after $d$ hours of delay relative to the original workload. These two transfer coefficients must satisfy the following constraints:

$$\begin{cases}\sum_{n}^{\mathrm{N}}x_n^{\mathrm{B,server}}=1\\ \sum_{t}^{24}\sum_{d}^{D_j}x_{j,t,d}^{\mathrm{B,time}}=1,\ \forall j\in J\end{cases} \tag{9}$$

### 2.2.2. Interactive workload transfer

The total amount of interactive workload received by the DC during

each time period of the day can be represented as:

$$L^{\mathrm{I}}_{24\times 1} = \left[l^{\mathrm{I}}_1, l^{\mathrm{I}}_2, \cdots l^{\mathrm{I}}_{24}\right] \tag{10}$$

where: $l^{\mathrm{I}}_t$ represents the total amount of interactive workload received by the DC at time $t$.

Since the number of interactive workload received at each moment is large and the energy required per task is relatively small, it is possible to split the interactive load received at each moment and distribute it across multiple servers for computation without delay. For the spatial transfer characteristics of interactive workload, the transferred interactive workload on each server can be represented as:

$$L^{\mathrm{I}} = \begin{bmatrix} x^{\mathrm{I}}_{1,1} l^{\mathrm{I}}_1 & \cdots & x^{\mathrm{I}}_{24,1} l^{\mathrm{I}}_{24} \\ \vdots & \ddots & \vdots \\ x^{\mathrm{I}}_{1,\mathrm{N}} l^{\mathrm{I}}_1 & \cdots & x^{\mathrm{I}}_{24,\mathrm{N}} l^{\mathrm{I}}_{24} \end{bmatrix} \tag{11}$$

where: $x^{\mathrm{I}}_{1,1}$ is the interactive workload spatial transfer coefficient, which is a constant between 0 and 1 and must satisfy the following constraints:

$$\sum_{n}^{N} x^{\mathrm{I}}_{t,n} = 1, \ \forall t \in [0, 24] \tag{12}$$

Thus, the expression for the load and the transferred load of a single server can be derived as:

$$\begin{cases} l_{t,n} = x^{\mathrm{B,server}}_{n} \sum_{j}^{J} \sum_{d}^{24} x^{\mathrm{B,time}}_{j,t,d} l^{\mathrm{B}}_{j,t} + x^{\mathrm{I}}_{t,n} l^{\mathrm{I}}_{t} \\ U_{t,n} = \dfrac{l_{t,n}}{F_{t,n}} \end{cases} \tag{13}$$

where:$l_{t,n}$ represents the total workload received by the $n$-th server at the time $t$; $U_{t,n}$ represents the load rate of the $n$-th server at time $t$; $F_{t,n}$ represent the maximum computational power available from the $n$-th server at time $t$, which is related to the server's core temperature $T^{\mathrm{CPU}}_{t,n}$; the rest of the symbols follow the definition in equations above.

And the energy consumption of servers is also related to their load rate and core temperature. The relationship between these variables can be expressed as[28][30]:

$$F_{t,n} = b_1 \left(T^{\mathrm{CPU}}_{t,n}\right)^3 + b_2 \left(T^{\mathrm{CPU}}_{t,n}\right)^2 + b_3 T^{\mathrm{CPU}}_{t,n} + b_4 \tag{14}$$

$$\begin{aligned} P^{\mathrm{server}}_{t,n} &= P^{\mathrm{idle}}_{n} + P^{\mathrm{dyn}}_{n,t} + P^{\mathrm{leak}}_{t} \\ &= P^{\mathrm{idle}}_{n} + \left(c_1 \left(U^{\mathrm{CPU}}_{t,n}\right)^3 + c_2 \left(U^{\mathrm{CPU}}_{t,n}\right)^2 + c_3 U^{\mathrm{CPU}}_{t,n}\right) + c_4 \exp\left(-\frac{c_5}{T^{\mathrm{CPU}}_{t,n}}\right) \end{aligned} \tag{15}$$

where: $b_1 \sim b_4$ are fitting coefficients used to characterize the relationship between the maximum computing power that the server can provide and the core temperature, which are computed mainly by referring to Ref. [30] and $c_1 \sim c_5$ are fitting coefficients used to characterize the relationship between the energy consumption of the server and the utilization rate and core temperature, which are computed mainly by referring to Ref. [28],$b_1 \sim b_4$ and $c_1 \sim c_5$ are fitted with the data sources in addition to the articles mentioned above, the SPEC public test dataset [18] and real DC operational data; $P^{\mathrm{idle}}_{n}$ represents the idle power of the $n$-th server, which is a constant; $P^{\mathrm{dyn}}_{n,t}$ represents the dynamic power of the $n$-th server which is related to the load rate at the time $t$; $P^{\mathrm{leak}}_{t}$ represents the leak power of the $n$-th server which is related to the core temperature at the time $t$.

Using the above methods, the total load and load rate for each server at each moment throughout the day can be calculated. And the transfer methods of the two types of workload can be illustrated as Fig. 3:

### 2.3. Heat flexibility

On the heat side, this paper mainly considers the adjustability of the

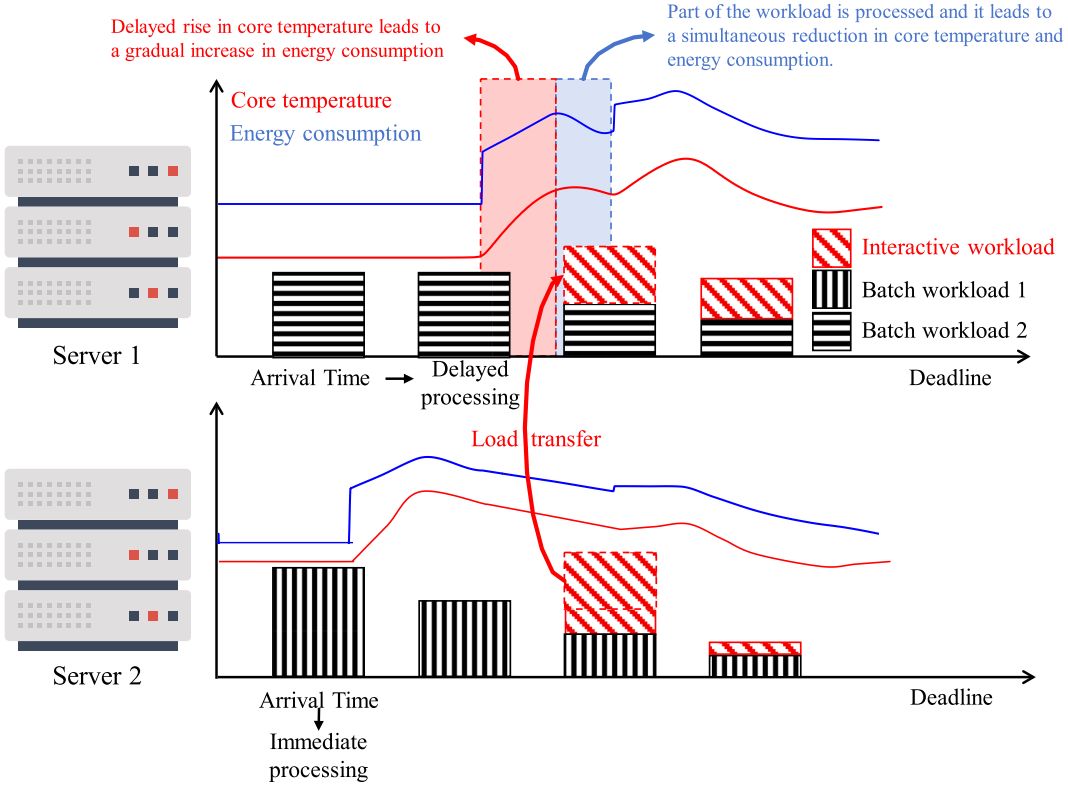



**Fig. 3.** The transfer methods of the two types of workload.

DC's inlet temperature and the resulting changes in the cooling system's energy consumption, as well as the virtual energy storage capacity caused by the building's thermal inertia [ [33]]. On the one hand, the energy consumption of the cooling system can be directly adjusted by regulating the inlet air temperature within an appropriate range; on the other hand, the cold load of the DC can be time-shifted to a certain extent through virtual energy storage. In particular, it can realize the flexible scheduling of DC electricity and cold load, and the specific DC heat flexibility carving and utilization are detailed in the following contents of this section.

Since DCs have high heat dissipation demands, maintaining the inlet temperature within a suitable range relies on the cooling system to provide cooling capacity, i.e., the room's cold load. The cold load does not immediately dissipate but rather goes through a slow decay process, causing the building to exhibit thermal storage characteristics. The relationship between the inlet temperature of the racks $T_t^{\mathrm{In}}$,the heat generation of the DC $H_t^{\mathrm{DC}}$, and the external heat dissipation$Q_t^{\mathrm{loss}}$ can be expressed as follows:

$$C^{\mathrm{DC}}\rho^{\mathrm{DC}}V^{\mathrm{DC}}\frac{dT_t^{\mathrm{In}}}{dt}=H_t^{\mathrm{DC}}-Q_t^{\mathrm{loss}} \tag{16}$$

$$\begin{aligned}Q_t^{\mathrm{loss}}&=Q_t^{\mathrm{cool}}+Q_t^{\mathrm{H1}}+Q_t^{\mathrm{H2}}+Q_t^{\mathrm{H3}}\\&=Q_t^{\mathrm{cool}}+(1+\eta)\frac{(1+\chi_1)(1+\chi_2)(1+\chi_3)S\alpha}{1/\theta_1+1/\theta_2+\sum\left(d_j/\sigma_j\right)}\left(T_t^{\mathrm{In}}-T_t^{\mathrm{W}}\right)\\&\quad+0.28C^{\mathrm{W}}\rho^{\mathrm{W}}kV^{\mathrm{W}}\left(T_t^{\mathrm{In}}-T_t^{\mathrm{W}}\right)\end{aligned} \tag{17}$$

where: $C^{\mathrm{DC}}$,$\rho^{\mathrm{DC}}$, and $V^{\mathrm{DC}}$ represent the specific heat capacity, density, and volume of the air inside the DC, respectively; $\eta$ represents the additional heat dissipation rate for cold air invasion through doors and windows in the envelope structure, with specific values typically referenced from design manuals; $\chi_1$, $\chi_2$ and $\chi_3$ represent additional heat dissipation correction parameters, specific reference to the "Code for design of heating ventilation and air conditioning"(GB50019–2003); $\theta_1$ and $\theta_2$ represent the heat transfer coefficients of the inner and outer surfaces of the structure (W/(m$^2$°C)); $d_j$ and $\sigma_j$ represent the thickness and thermal conductivity of each layer of the envelope structure; k represent the daily heat exchange frequency of DC; $C^{\mathrm{W}}$, $\rho^{\mathrm{W}}$, and $V^{\mathrm{W}}$ represent the specific heat capacity, density, and volume of the external infiltrating air, respectively; $Q_t^{\mathrm{cool}}$ is the output of the cooling system; $Q_t^{\mathrm{H1}}$, $Q_t^{\mathrm{H2}}$, and $Q_t^{\mathrm{H3}}$ represent the heat transfer dissipation through the enclosure structure, the cooling air infiltration dissipation, and the cooling air intrusion dissipation, respectively; $S$ is the external heat transfer area of the DC's enclosure structure; $T_t^{\mathrm{W}}$ is the external temperature; The factor 0.28 is a unit conversion constant, where 1 kJ/h equals to 0.28 W.

From Eq. (13), it can be calculated that for the hourly-level scheduling performed in this paper, this timescale is much larger than the time required for the cooling system's air supply temperature to cause changes in inlet temperature of the racks and reach thermal equilibrium. Therefore, it is assumed that within the scheduling period, the inlet temperature of the racks remains constant and is the same as the air supply temperature from the air conditioning system, both being $T_t^{\mathrm{In}}$. To simplify the calculation, it is assumed that the return air temperature is consistently 32 °C. At the same time, for DC operation reliability requirements, we refer to the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) recommended air supply temperature range, set the DC air supply temperature should always be maintained at 18 to 27 °C [34].

At the same time, the COP of the cooling system is defined as the ratio of the room's cold load to the energy consumption of the cooling system. For traditional air-cooled cooling systems, the COP is closely related to $T_t^{\mathrm{In}}$. Thus, adjusting the target temperature will not only affect the server's operating state and power consumption but also have an impact on the cooling system's energy consumption. The specific impact mechanism is shown as follows:

$$K_t^{\mathrm{COP}}=k_1\left(T_t^{\mathrm{In}}\right)^2+k_2T_t^{\mathrm{In}}+k_3 \tag{18}$$

$$P_t^{\mathrm{cool}}=\frac{Q_t^{\mathrm{cool}}}{K_t^{\mathrm{COP}}}=\frac{H_t^{\mathrm{DC}}-\sum_{i=1}^{3}Q_t^{Hi}-C^{\mathrm{DC}}\rho^{\mathrm{DC}}V^{\mathrm{DC}}\frac{dT_t^{\mathrm{In}}}{dt}}{k_1\left(T_t^{\mathrm{In}}\right)^2+k_2T_t^{\mathrm{In}}+k_3} \tag{19}$$

where: $K_{\mathrm{COP},t}$ is the COP of the cooling system at time $t$; $k_1$~$k_3$ are fitting coefficients [35]; $P_t^{\mathrm{cool}}$ is the energy consumption of the cooling system at time $t$.

Sections 2.1 to 2.3 focus on the individual operation and flexibility resource utilization models for the computing system (servers), power supply system, and cooling system. On this basis, there is a close coupling relationship between the three systems. From the perspective of energy consumption, the servers in the DC consume electrical energy, generating an equivalent amount of heat. This thermal energy is dissipated by the cooling system through the temperature difference between the DC and the outside environment. During this process, the cooling system also consumes electrical energy. The electrical consumption of both the servers and the cooling system together forms the electricity load of the DC, which is supplied by the power system:

$$P_t^{\mathrm{DC}}=\sum_n^N P_{n,t}^{\mathrm{server}}+P_t^{\mathrm{cool}} \tag{20}$$

$$H_t^{\mathrm{DC}}=\sum_n^N P_{n,t}^{\mathrm{server}} \tag{21}$$

Additionally, the cooling system can also affect the operating state of the servers by changing the temperature. Specifically, the core temperature of the server is related to both the air supply temperature and the load rate [25]:

$$T_{t,n}^{\mathrm{CPU}}=T_t^{\mathrm{In}}+a_1\left(1-e^{-a_2U_{t,n}}\right) \tag{22}$$

where $a_1$ and $a_2$ are fitting parameters.

Based on this, a comprehensive flexibility resource collaborative scheduling framework for the DC is constructed: the computing, electricity, and cooling systems are closely interconnected through the conversion and supply of energy. Additionally, the computing system and the cooling system are further coupled through temperature regulation. In the variables, the workload and the external temperature are external variables, which change over time and cannot be controlled. All other variables related to the computing side and heat side can be adjusted through workload transfer and air supply temperature scheduling. The electricity side mainly considers the hourly output of the three types of power supply equipment. For DCs, one of the main optimization objectives is to complete as much workload as possible with minimal resources. Therefore, the computing efficiency index is selected for preliminary research on the regulation capability of load rate and air supply temperature for the system. The server computing efficiency and the overall computing efficiency of the DC under different load rates and air supply temperatures are calculated based on the relationships above. The relationships between load rate, air supply temperature, core temperature, and other variables are detailed in Appendix A.

By comparing Fig. 4(a) and 4(b), it can be seen that, when the cooling system's energy consumption is not considered, the computing efficiency of a single server is highest when the inlet temperature is at its lowest (18 °C). However, when considering the cooling system's energy consumption, the maximum computing efficiency for the entire DC

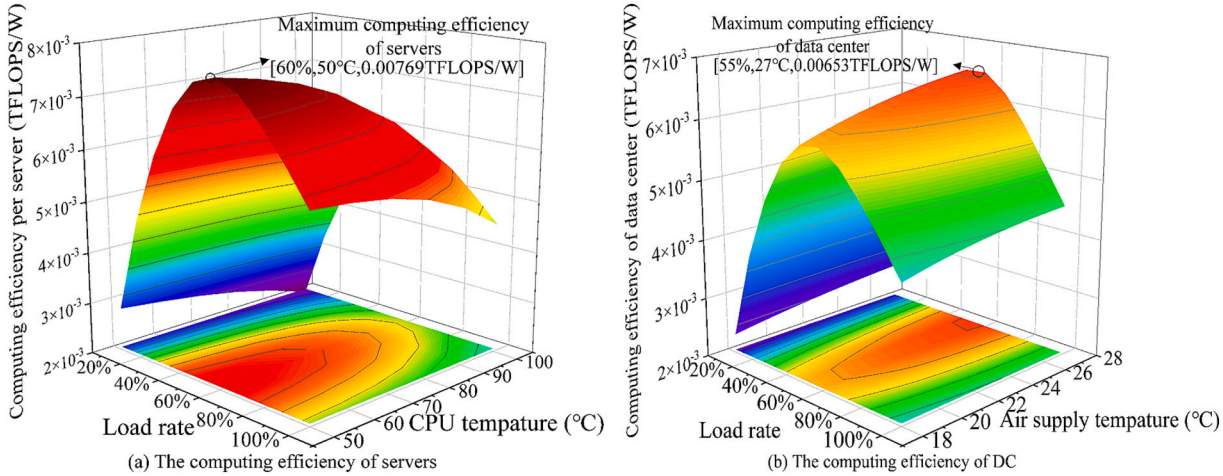



**Fig. 4.** Computing efficiency under different load rates and air supply temperature.

(assuming all servers have the same load rate) occurs at a higher inlet temperature (27 °C). This indicates that the coupling mechanism between computing and cooling systems has a significant impact on DC planning and scheduling. To comprehensively consider the energy consumption of the DC, whether or not to account for the synergy between cooling and electricity supply systems in the optimization scheduling results is crucial, and the collaborative mechanism of all-element flexibility resources should be taken into account.

## 3. Optimization method

### 3.1. Optimization objectives and constraints

The economic objective mainly includes investment costs for the energy supply devices and energy purchase costs, as shown in Eq. (23):

$$F_1 = \min\left(\sum_i \frac{c_i^{\text{inv}} P_i^{\text{c}} r(1+r)y^{\text{run}}}{365.25((1+r)y^{\text{run}}-1)} + \sum_t \left(c_t^{\text{grid,p}} P_t^{\text{grid}} + c_t^{\text{gas,p}} f_t^{\text{gt}}\right)\right) \tag{23}$$

where: $c_i^{\text{inv}}$ is the unit installation cost of equipment $i$; $P_i^{\text{c}}$ is the installed capacity of equipment $i$; r is the interest rate; $y^{\text{run}}$ is the DC's run year; $c_t^{\text{grid,p}}$ and $c_t^{\text{gas,p}}$ are the time-of-use grid price and natural gas price, respectively.

The carbon emission objective mainly considers electricity supply from the grid and the purchase of natural gas, as shown in Eq. (24):

$$F_2 = \min\left(\sum_t \left(c_t^{\text{grid,c}} P_t^{\text{grid}} + c^{\text{gas,c}} f_t^{\text{gt}}\right)\right) \tag{24}$$

where: $c_t^{\text{grid,c}}$ $c_i^{\text{inv}}$and $c^{\text{gas,c}}$ are the carbon emission factors for the grid and natural gas, respectively, in units of kg/kWh.

The energy consumption objective mainly includes server energy consumption and cooling system energy consumption, as shown in Eq. (25):

$$F_3 = \min\left(\sum_t \left(\sum_n^N P_{t,n}^{\text{server}} + P_t^{\text{cool}}\right)\right) \tag{25}$$

At the same time, PUE and computing efficiency $\lambda^{DC}$ are used as evaluation metrics. It is assumed that servers and cooling systems account for about 90 % of the total energy consumption of the DC. The relationship between the two is shown below:

$$\begin{cases} PUE = \dfrac{\left(\sum_t \left(\sum_n^N P_{t,n}^{\text{server}} + P_t^{\text{cool}}\right)\right)\Big/0.9}{\sum_t \sum_n^N P_{t,n}^{\text{server}}} \\ \lambda^{DC} = \dfrac{\sum_t \sum_n^N l_{t,n}}{\left(\sum_t \left(\sum_n^N P_{t,n}^{\text{server}} + P_t^{\text{cool}}\right)\right)\Big/0.9} \end{cases} \tag{26}$$

Since the total workload of the DC is set as constant in all the scenarios, the computing efficiency of the DC depends solely on energy consumption, while PUE is only related to the energy consumption proportion of the cooling system. Therefore, PUE and computing efficiency are not optimized independently but are analyzed as an evaluation index related to the optimization objective of minimizing energy consumption.

Combined with Section 2, the overall optimization model proposed in this paper is shown below:

$$\begin{cases} \min f = \sum_{i=1}^{3} \omega_i \dfrac{F_i - F_{i,\min}}{F_{i,\max} - F_{i,\min}} \\ \text{s.t.} \begin{cases} P_t^{\text{DC}} = P_t^{\text{grid}} + P_t^{\text{gt}} + P_t^{\text{ba,out}} - P_t^{\text{ba,in}} \\ H_t^{\text{DC}} = C^{\text{DC}} \rho^{\text{DC}} V^{\text{DC}} \dfrac{dT_t^{\text{In}}}{dt} + Q_t^{\text{loss}} \\ P_t^j = \underset{x \in \phi_{p,j}}{g_{p,j}}(x); H_t^k = \underset{x \in \phi_{h,k}}{g_{h,k}}(x); Q_t^l = \underset{x \in \phi_{q,l}}{g_{q,l}}(x) \\ j \in \psi_p; k \in \psi_h; l \in \psi_q \end{cases} \end{cases} \tag{27}$$

where: $f$ represents the final optimization objective, for single-objective optimization, $f$ equals to one of $F_1$, $F_2$, $F_3$, for multi-objective optimization, $f$ is equal to the weighted sum of $F_1$, $F_2$, $F_3$ after normalization, $\omega_i$ is the objective weight, which is always greater than or equal to 0 and sums up to 1; $F_{i,\max}$ and $F_{i,\min}$ are the maximum and minimum values of the optimization objective $F_i$, which are obtained by sole-objective optimization, for example, $F_{1,\min}$ is the optimization result when the single optimization objective is $F_1$, and $F_{1,\max}$ is the larger value of the $F_1$ result obtained when the single optimization objective is $F_2$ or $F_3$; the optimization constraints include, from top to bottom, the balance of electricity power, the balance of cooling/heating energy, and the constraints of each equipment's output/load, of which $g_{p,j}$, $g_{h,k}$, and $g_{q,l}$ are the equipment's output/load functions of $j$, $k$, $l$, respectively. See Section

2 for details, $x$ denotes the factors affecting the equipment output/load, such as load rate, room temperature, etc., which are constrained by the power, heat, and cooling constraint sets $\Phi_{p,j}$, $\Phi_{h,k}$, $\Phi_{q,l}$, see Section 2 for details; and $\Psi_p$, $\Psi_h$, $\Psi_q$ are the sets of electricity/heat/cold-producing/consuming equipment.

### 3.2. Optimization framework

Section 2 discusses the distribution of computing efficiency in the DC, which is only applicable to the flexibility resource scheduling at a single time slot. It has not yet considered the time distribution of electricity-side flexibility resources, nor the time transfer characteristics of workload and virtual energy storage. To address this, this paper proposes an overall framework that comprehensively considers the computing-heat coupling mechanism in the DC and performs multi-objective optimization using the DC's own flexibility resources, as shown in Fig. 5:

The framework is divided into four parts and establishes a two-stage optimization model to solve the DC flexibility resource utilization planning and scheduling scheme. The first part, combined with the actual DC operation data, fits or determines each parameter and establishes a complete DC operation scheduling and flexibility resource utilization model according to the content of Chapter 2; the second part, i.e., the upper load-side optimization scheduling, according to the determination of the optimization objective, schedules the computing power load and temperature, and then determines the energy load of the DC, which is passed to the lower energy supply-side optimization scheduling model; the third part: Planning and scheduling of the installed capacity and output of various types of energy supply equipment according to the energy load passed to the upper layer. The load side and the energy supply side are closely integrated, and the optimization scheduling problem is a typical mixed-integer quadratic constrained programming problem, which is calculated by one-step solution using the Gurobi solver. This method is widely used in solving the integrated energy system planning and scheduling. Integrated energy system planning and scheduling problems [36–38]. In the fourth section, the optimized scheduling results are analyzed and sensitivity analyses are performed on key influencing factors such as arithmetic load types, time-sharing tariffs, renewable energy supply, and virtual energy storage capacity of the DC, in order to more deeply explore the DC flexibility resources and utilization path.

Based on the actual operation and load data of DCs in Beijing, China, this paper designs the flexibility utilization model of a typical DC, while considering the influence of external factors such as ambient temperature and grid time-sharing tariffs on DCs in Beijing, and the baseline PUE of this DC is set at 1.37 considering local climates of Beijing, China. And the main parameters of the DC studied are listed in Appendix B.

## 4. Case study

### 4.1. Analysis of the impact of flexibility resources on reducing carbon emissions, energy consumption and economic cost

For the optimization objectives mentioned in Section 3.1, single-objective optimization is performed first, and the types of flexibility resources used in the DC are combined, leading to 8 scenarios: S1: No flexibility resources are used at all, i.e., no workload transfer, full reliance on grid power, and the cooling system always operates at the minimum temperature (18 °C); S2: Only heat flexibility resources are considered; S3: Only electricity-side flexibility resources are considered; S4: Both heat and electricity-side flexibility resources are considered; S5: Only computing-side flexibility resources are considered; S6: Both computing-side and heat flexibility resources are considered; S7: Both computing-side and electricity-side flexibility resources are considered; S8: All three types of flexibility resources are utilized simultaneously, i.e., workload transfer, flexible switching of energy supply modes, and free adjustment of target temperature within the allowable temperature range (18–27 °C). The scenarios settings are organized in Table 2.

Each of the 8 scenarios is optimized with the three objectives, and the calculation results are provided in Appendix C.

Using S1 as the baseline scenario and its optimization results as the benchmark results, by comparing the optimization results of the other seven types of scenarios with it, we are able to calculate the impacts of the utilization of different types of flexibility resources and their coupling effects on the DC's energy consumption, economic cost, and carbon emission in terms of energy-saving rates, carbon saving rates, and cost reduction ratios, as shown below:

$$\begin{cases} r_{i,j} = \dfrac{F_{i,1} - F_{i,j}}{F_{i,1}} \\ i \in \{1,2,3\} \\ j \in \{2,3,4,5,6,7,8\} \end{cases} \tag{28}$$

where: $Fi,j$ denotes the optimization result of optimization objective $i$ under the $j$-th scenario, and $Fi,1$ denotes the optimization result of optimization objective i under S1, which is the baseline value; $r1,j$, $r2,j$, $r3,j$ denote the energy-saving rate, the carbon saving rate, and the proportion of cost reduction of the DC under the $j$-th scenario, respectively. From this, the impact of flexibility resource utilization on DC energy saving, emission reduction and sustainable development can be quantitatively analyzed.

Based on the optimization calculation results in Appendix C, the impact of flexibility resource utilization on reducing the overall carbon emissions, energy consumption, and economic costs of the DC is analyzed, as shown in Fig. 6:

It can be observed from Fig. 6 that different flexibility resources have varying impacts on the overall carbon emissions, energy consumption, and economic costs of the DC, along with their internal mechanisms. Under the DC flexibility resource utilization framework designed in this paper, the reasonable scheduling of flexibility resources can reduce carbon emissions by up to 38.5 %, energy consumption by up to 22.8 %, and economic costs by up to 21.6 %.

From the perspective of the underlying mechanisms, there are significant differences in the effects of the three types of flexibility resources of DC. As shown in Fig. 6(b), the utilization of electricity-side flexibility resources does not directly reduce the total energy consumption of the DC, while computing-side and heat-side flexibility resources effectively reduce the energy consumption of servers and the cooling system by adjusting the load rate and air supply temperature, respectively, thereby indirectly reducing economic costs and carbon emissions (mainly from grid electricity supply).

For example, in Fig. 6(b), using both computing-side and heat-side flexibility resources results in a larger reduction in the total energy consumption of the DC (22.8 %) compared to using only computing-side flexibility resources (11.1 %) or heat-side flexibility resources (13.1 %). However, this reduction is less than the sum of the two individual effects, with the overlap accounting for approximately 13.5 % of the energy savings from computing-side flexibility resources and 11.5 % from heat-side flexibility resources. The overlap mainly arises from the need to increase the air supply temperature to reduce the energy consumption of the cooling system, which inevitably leads to an increase in server energy consumption. Similar conclusions can be drawn from Fig. 6(a) and 6(c).In order to calculate the size of the overlap part, i.e., the part of different flexibilities that affect or constrain each other, also taking Fig. 6(b) as an example, the optimization objectives of S1, S2, S5 and S6 are calculated with the lowest energy consumption as the optimization objective, and it is obtained that compared to S1, the energy consumption under S2, S5 and S6 can be reduced by 13.1 %, 11.1 % and 22.7 %, respectively. The energy consumption reduction under S2 and S5 can be added up to 24.2 %, which is greater than the 22.7 % of S6, and the difference can be regarded as the overlap part (1.5 %) due to the

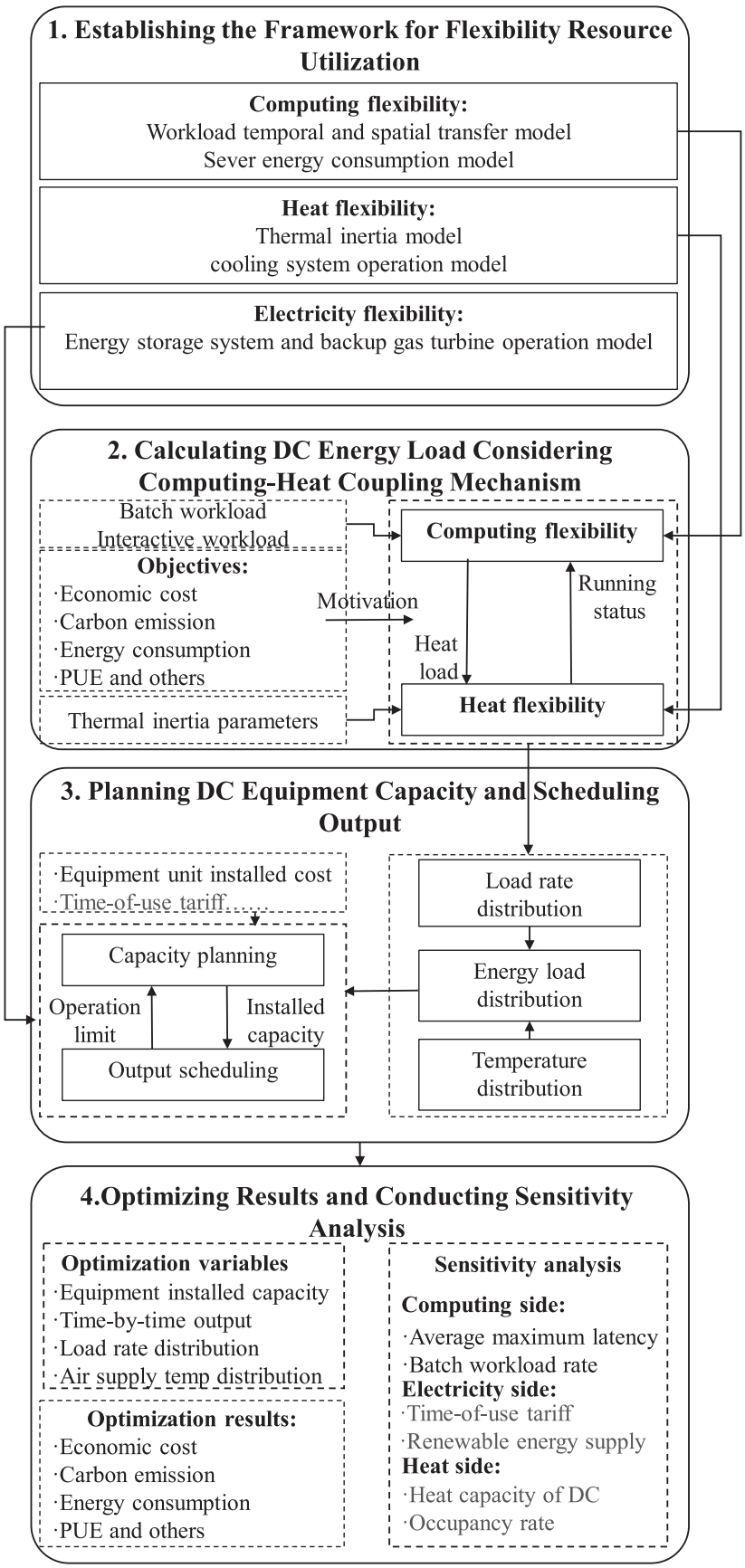



**Fig. 5.** Overall framework considering the computing-heat coupling mechanism.

**Table 2**
The scenarios settings.

| Scenarios | Electricity side | Computing side | Heat side |
|---|---|---|---|
| S1 | Only grid power supply | No workload transfer | Fixed temperature setting |
| S2 | Only grid power supply | No workload transfer | Optimized temperature setting |
| S3 | Backup power supply used | No workload transfer | Fixed temperature setting |
| S4 | Backup power supply used | No workload transfer | Optimized temperature setting |
| S5 | Only grid power supply | Optimized workload transfer | Fixed temperature setting |
| S6 | Only grid power supply | Optimized workload transfer | Optimized temperature setting |
| S7 | Backup power supply used | Optimized workload transfer | Fixed temperature setting |
| S8 | Backup power supply used | Optimized workload transfer | Optimized temperature setting |

interaction of two types of flexibilities. The other cases are calculated in the same way, and the correctness of this calculation method is verified by comparing the optimization results of the cases with only one type of flexibility (S2, S3, and S5) with the optimization results of the cases with two types of flexibilities (S4, S6, and S7), obtaining the part of the interactions between the various types of flexibilities and the part of their individual effects, all of which, when summed together, are equal to the optimization result of S8.

The mechanism of action for electricity-side flexibility resources mainly relies on the time-of-use tariffs and the differences in carbon emission factors for grid and natural gas. By flexibly scheduling the supply mode and battery charge/discharge at different times, carbon emissions or economic costs can be effectively reduced. Therefore, in Fig. 6(c), the reduction in economic costs from utilizing electricity-side flexibility resources overlaps only slightly with the utilization of computing-side flexibility resources. This overlap is related to the time transfer of workloads and the time distribution of supply costs. In Fig. 6 (a), the reduction in carbon emissions from electricity-side flexibility resources overlaps with the effects of the other two types of flexibility resources, due to the commonality of two emission reduction methods: "reducing grid electricity supply to lower carbon emissions" and "using lower-carbon-emission gas turbines to partially replace grid electricity supply."

Further, Fig. 7, the radar charts, illustrates the capabilities of the three types of flexibility resources in terms of energy saving, carbon reduction, cost reduction, reducing PUE, and improving computing efficiency, as shown in the figure. Based on the types of flexibility resources utilized, the cases are divided into S1 to S8, with S1 being the baseline case, where no flexibility resources are used. The different colors in the radar chart represent the corresponding optimization objectives, and negative values indicate that utilizing these flexibility resources may have an opposite effect on the objective under that specific scenario or optimization target. For example, in case S3, where only electricity-side flexibility resources are used, the energy consumption reduction is always 0 %. When carbon emissions are the optimization target, S3 can reduce carbon emissions by 21.5 %, but it simultaneously increases economic costs by 14.4 %, as shown in the chart, which equals a reduction of −14.4 % in economic costs.

From this, the different characteristics of the three types of flexibility resources can be further identified: While electricity-side flexibility resources do not have energy-saving capabilities, they are highly effective in reducing costs and carbon emissions. They can balance the trade-off between cost reduction and carbon reduction by adjusting the supply mode. Heat and computing-side flexibility resources mainly achieve cost and carbon reduction through energy savings. In fact, both can implement a certain degree of load regulation through virtual energy storage and time transfer of workload. However, considering the large cold load

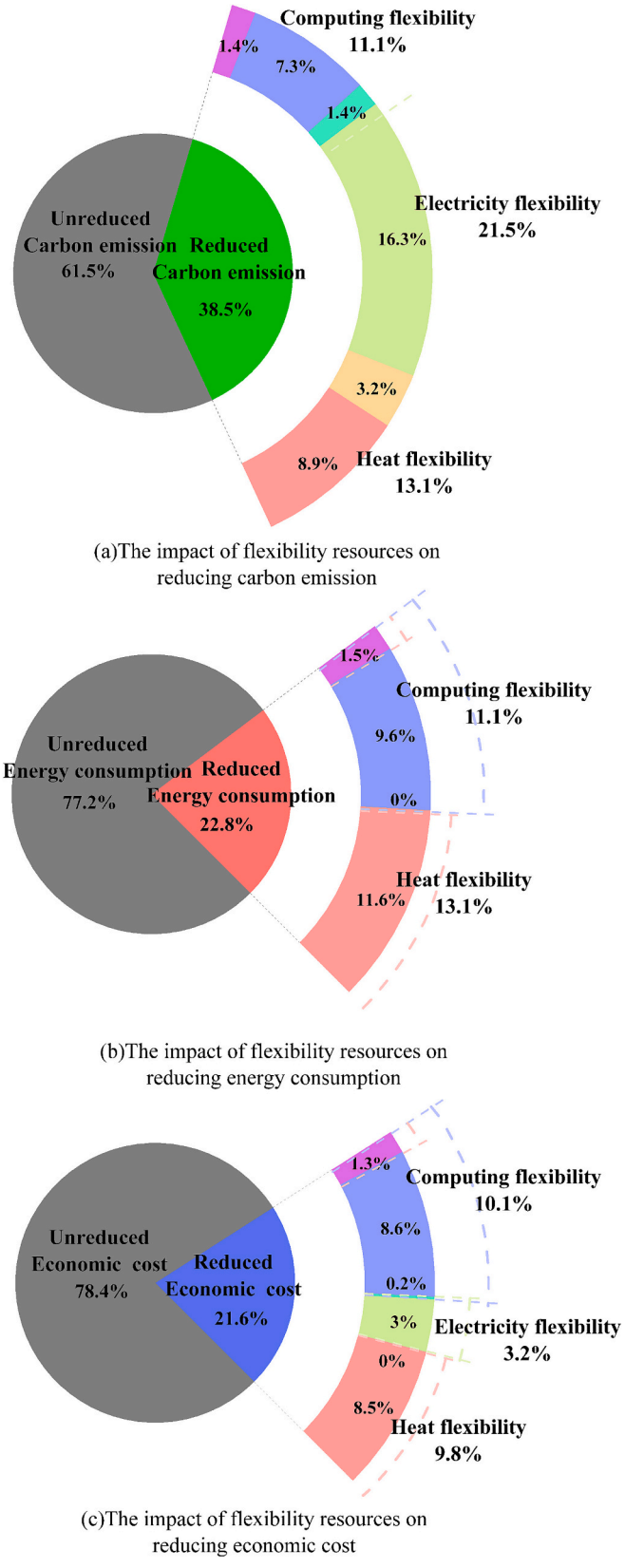



**Fig. 6.** The impact of flexibility resource utilization on DC.

of DCs, the load adjustment capacity of virtual energy storage is limited, while the time transfer of workload has a more significant effect. Furthermore, under the condition of using only grid electricity, the time transfer effect is still relatively small. Considering PUE and computing

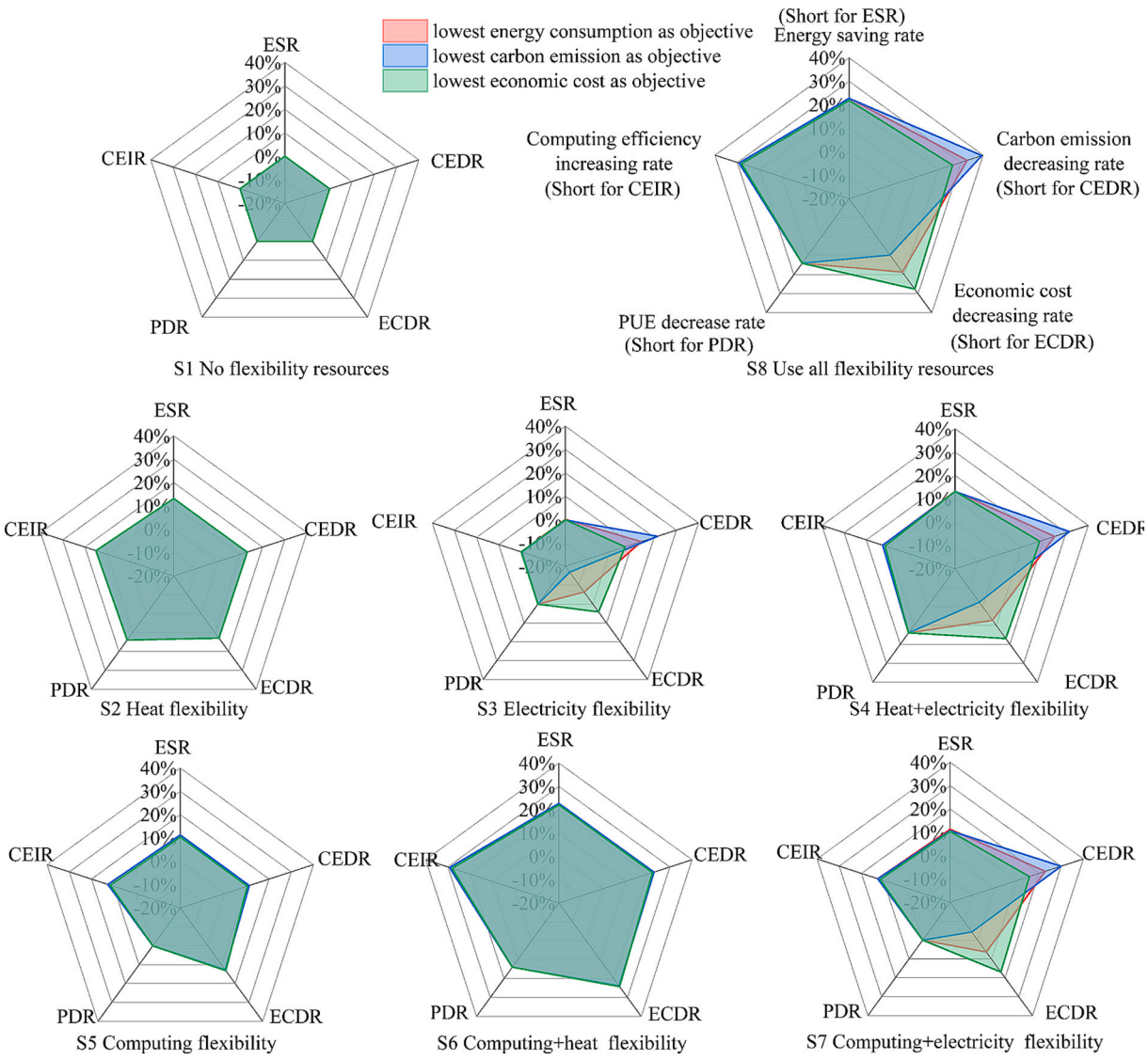


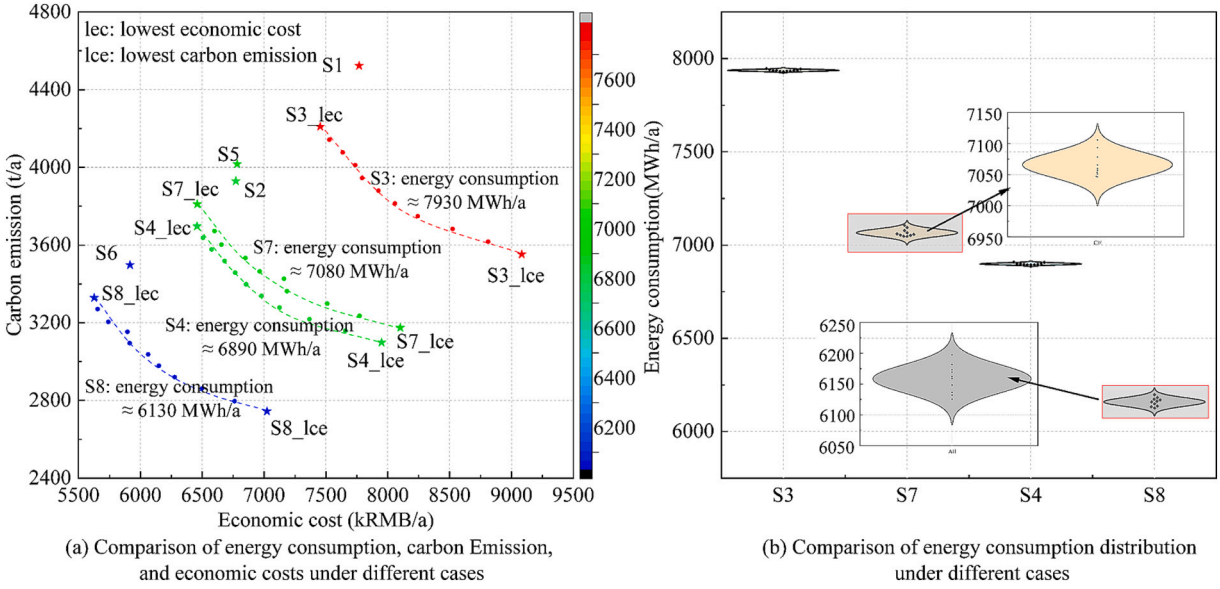

**Fig. 7.** The capabilities of the flexibility resources in terms of optimization objectives.

efficiency, in different scenarios, the total amount and distribution of workload received by the DC are the same, so both PUE and computing efficiency can be considered as energy consumption-related metrics. The former is mainly related to cooling system energy consumption. With fixed server energy consumption, the lower the cooling system energy consumption, the lower the PUE. Therefore, in Fig. 8, PUE shows significant improvement in the four cases (S2, S4, S6, S8) that consider heat flexibility resources, dropping from 1.37 to 1.18. The computing efficiency metric is related to the overall energy consumption of the DC, and the lower the energy consumption, the higher the computing efficiency, exhibiting an inverse relationship. Its change mechanism is consistent with the energy consumption variation mechanism mentioned above.

To further explore the synergistic effects between the three types of flexibility resources, the energy consumption, costs, and carbon emission data for all the calculated cases are shown in Fig. 8. To demonstrate the adjustment capabilities of electricity-side flexibility resources, the pareto curve of carbon emissions and costs for the cases utilizing electricity-side flexibility resources is constructed using the weighting method. Fig. 8(b) shows the energy consumption distribution for the four cases that include electricity-side flexibility resources. It can be seen that the energy consumption distribution for cases S7 and S8, which include computing-side flexibility resources, is more spread out, indicating that when electricity-side and computing-side flexibility resources work together, computing-side flexibility not only reduces the energy consumption per computational power through spatial load transfer but also slightly increases the energy consumption through time load transfer in order to achieve better cost and carbon reduction effects. This further emphasizes the importance of the synergistic effects between flexibility resources in achieving overall optimization.

The server load rate distribution for different scenarios is shown in Fig.9. The horizontal width of the violin plot represents the load rate distribution density, while the vertical length represents the distribution range. From the perspective of computing-side flexibility resource utilization, compared to schemes that do not consider computing-side flexibility, the schemes that include computing-side flexibility concentrate the workload as much as possible around the “optimal load rate corresponding to unit energy consumption.” Simulation results show that utilizing workload flexibility can reduce energy consumption by up to 11.1 %, while also reducing carbon emissions and economic costs for the system.

Further comparison of the economic optimal solution and carbon emission optimal solution reveals that in the carbon emission optimal solution, the server load rate distribution is more dispersed, and the air supply temperature remains constant. This is due to the synergistic effects between electricity-side flexibility resources and computing-side and heat-side flexibility resources. In the carbon emission optimal




(a) Comparison of energy consumption, carbon Emission, and economic costs under different cases

(b) Comparison of energy consumption distribution under different cases

**Fig. 8.** The optimization results for all the calculated cases.

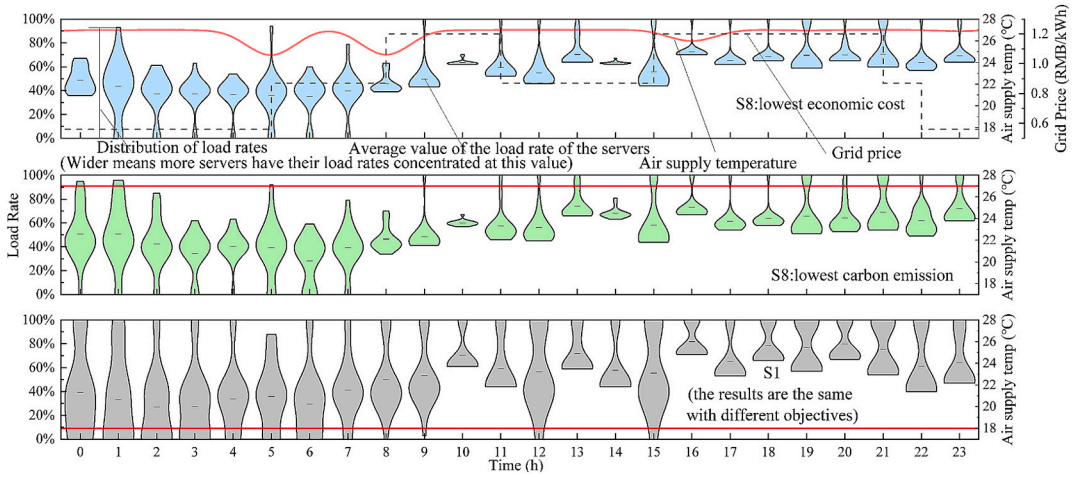



**Fig. 9.** The server load rate distribution for different scenarios.

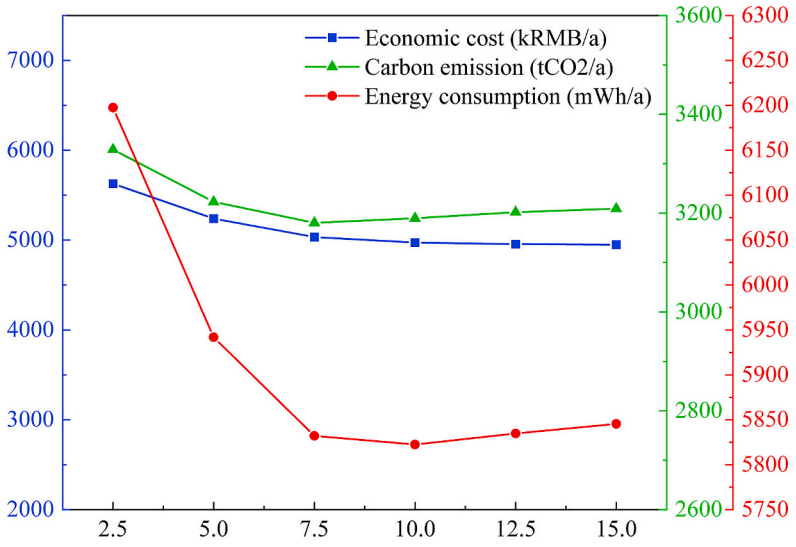
solution, the DC is fully powered by gas turbines, and the carbon emission per unit of energy does not show significant variation over time. At the same time, due to the limited momentum of temporal transfer for workload and virtual energy storage, the amount of transfer is also limited, resulting in a more dispersed server load rate distribution, while the air supply temperature remains at the lowest overall energy consumption point for the DC.

To generalize the results of this study and further explore the utilization of DC flexibility resources under different scenarios, a sensitivity analysis was conducted on three key factors influencing the utilization of flexibility resources in DCs. These three factors are: the delay time of batch workload, the electricity price of the grid, and the thermal inertia of the DC building. All cases were calculated based on the consideration of all three flexibility resources, with the optimization objective being economic optimization. A sensitivity analysis of the impact of the utilization of computational flexibility (changing the average maximum latency and batch workload rate), power flexibility (changing the average grid price and introducing wind and solar power), and heat flexibility (changing the number of racks without changing the number of servers) is performed while controlling for the other data center parameters to be consistent with Section 4.1 (see Appendix B for more details on the data) and the results are shown in Sections 4.2–4.4.

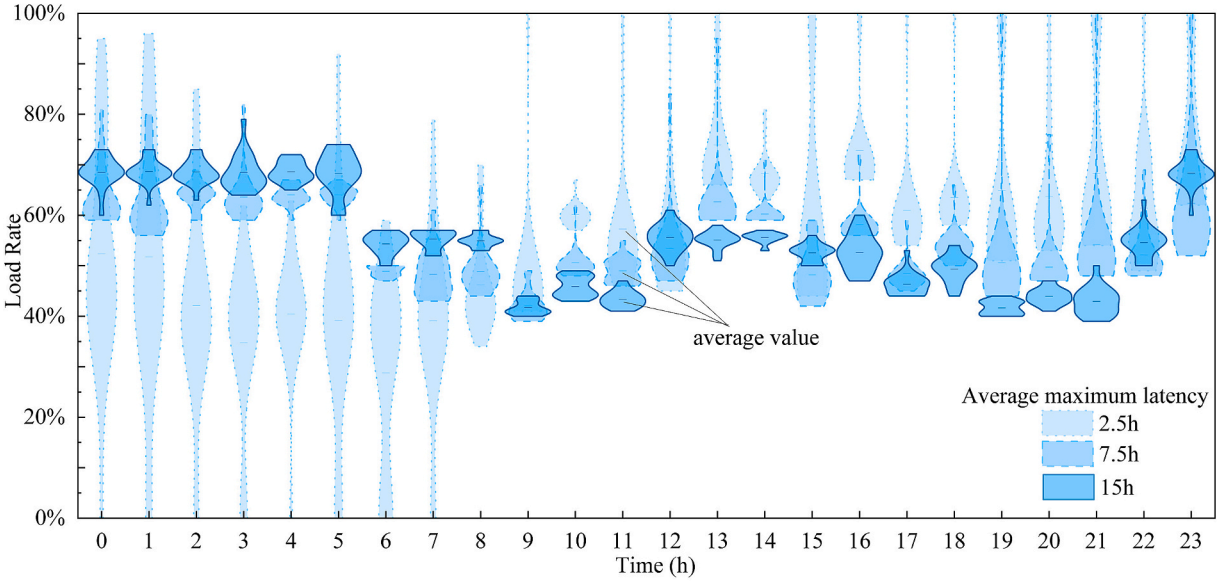
### 4.2. *Sensitivity analysis of computing flexibility*

In the original scenario, the average maximum delay time or latency for the DC's batch workload was set to 2.5 h. To explore the impact of this factor on the utilization of flexibility resources in the DC, this paper proportionally increases the maximum delay time of each batch workload and performs calculations. The results are shown in Fig. 10(a). Further, cases with average maximum delay times of 2.5 h, 7.5 h, and 15 h were selected, and the distribution of server load rates for each case was analyzed, as shown in Fig. 10(b).

In the initial stages, the energy consumption, carbon emissions, and economic costs of the DC all show a downward trend. During this phase, the main objective of workload transfer is to reduce energy consumption per computational power. When the average maximum delay time reaches 7.5 to 10 h, the transferable workload can keep most of the server load rates close to the load rate at which CECPCP is minimized. At this point, the driving force of time-of-use electricity prices gradually exceeds the driving force to reduce energy consumption. As the maximum delay time increases further, the workload distribution begins




(a) The economic cost, carbon emission and energy consumption changes by the average maximum latency




(b) The load rate distribution changes by the average maximum latency

**Fig. 10.** The impact of average maximum latency on the utilization of flexibility resources.

to show an inverse trend with the time-of-use electricity price distribution, i.e., during high electricity price periods, the average load rate of the servers is lower. This phenomenon suggests that when the computing workload in the DC has a high degree of delay tolerance, the trend of load transfer is strongly correlated with electricity-side flexibility resources.

At the same time, the type of computing workload received by the DC also significantly impacts energy consumption and has a clear coupling relationship with the average maximum latency of batch workload. Based on the above study, this paper also considers the workload types and their proportions, using energy consumption minimization as the optimization objective, to explore how load transfer can minimize energy consumption. The results are shown in Fig. 11:

When the average maximum latency of batch workload increases, a higher proportion of batch workload can effectively reduce the energy consumption of the DC. When the average maximum latency exceeds 12.5 h and the proportion of interactive workload reaches 70 %, the overall energy consumption of the DC decreases from 6130 MWh per year to 5816 MWh per year, reducing energy consumption by an additional 6 %.

### 4.3. Sensitivity analysis of electricity flexibility

In the original scenario, the average grid price was approximately 0.9 RMB/kWh. To explore the impact of grid prices on the utilization of flexibility resources in DCs, this paper scales the average grid price either down or up by increasing or decreasing the grid price in equal proportions for each time of day and performs calculations. The results are shown in Fig. 12(a). Additionally, Fig. 12(b) shows the changes in the daily average output of devices at different average grid prices.

On the supply side, the grid and gas turbines substitute each other. When the average electricity price of the grid is less than or equal to 0.72 RMB/kWh, the economic cost of using gas turbines for power supply exceeds the highest electricity price, so at this point, the DC is entirely

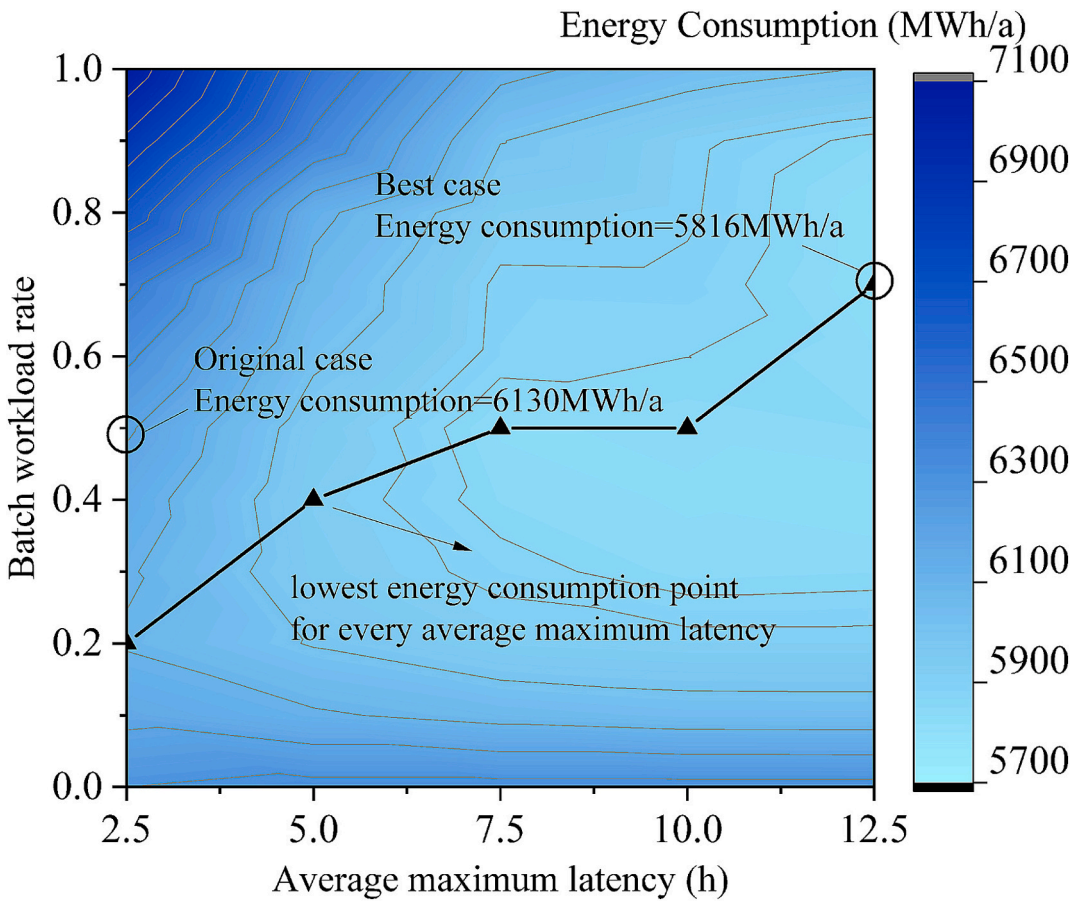



**Fig. 11.** The impact of average maximum latency and batch workload rate.

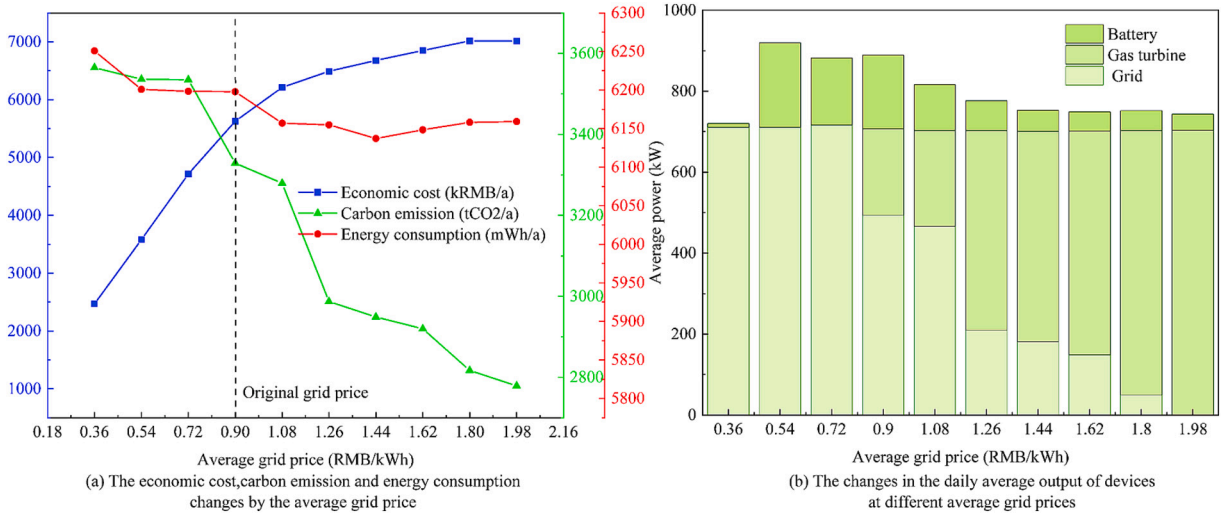



**Fig. 12.** The impact of time-of-use grid price on the utilization of flexibility resources.

powered by the grid. As the electricity price increases, the proportion of gas turbine output gradually increases, until the average grid price exceeds 1.98 RMB/kWh, at which point the gas turbine's economics becomes entirely superior to grid power supply.

From the flexibility perspective, when the grid electricity price is too low, it fails to fully stimulate the utilization of computing-side and electricity-side flexibility resources. Only when there is a certain difference between the peak and off-peak electricity prices does the energy storage device start to play a role. It charges during the low-price periods and discharges during the high-price periods. Similarly, when the average grid price exceeds 0.72 RMB/kWh, the peak-to-valley price difference will start to motivate the workload to shift to the low-price periods. As the output of the gas turbine increases, the electricity costs across different periods gradually converge until they become equal, at which point the battery output decreases, and the main driver for workload transfer becomes the overall reduction in energy

consumption.

Meanwhile, in view of the demand for sustainable development of DCs, in recent years, many DCs have included peripheral renewable resources into the construction of power supply systems during the construction process [39,40]. Therefore, this paper takes the installed capacity of renewable energy power supply equipment represented by wind power (WP) and photovoltaic power (PV) generation as variables, and explores the impact of the utilization of flexible resources on the wind and light consumption ability of DCs on the one hand, and the impact of the utilization of renewable resources on the carbon emission, energy consumption and economic cost of DCs on the other. The new energy balance constraints of the system and the operating constraints of the additional equipment when considering the addition of WP and PV equipment are shown below:

$$\begin{cases} P_t^{\mathrm{DC}} = P_t^{\mathrm{grid}} + P_t^{\mathrm{gt}} + P_t^{\mathrm{ba,out}} - P_t^{\mathrm{ba,in}} \\ \qquad +P_t^{\mathrm{wp}} + P_t^{\mathrm{pv}} - P_t^{\mathrm{re,loss}} \\ P_t^{\mathrm{wp}} = \eta_t^{\mathrm{wp}} P^{\mathrm{c,wp}} \\ P_t^{\mathrm{pv}} = \eta_t^{\mathrm{pv}} P^{\mathrm{c,pv}} \\ 0 \le \sum_t P_t^{\mathrm{re,loss}} \le 0.05 \sum_t \left(P_t^{\mathrm{pv}} + P_t^{\mathrm{wp}}\right) \end{cases} \tag{29}$$

where: $P_t^{\mathrm{wp}}$ and $P_t^{\mathrm{pv}}$ represent the output of WP and PV, respectively; $\eta_t^{\mathrm{wp}}$ and $\eta_t^{\mathrm{pv}}$ represent the unit capacity output of WP and PV, respectively; represents allowable abandoned renewable power which should be less than 5 % of the total output of WP and PV. The investment cost design is mainly based on the Ref. [41]

As shown in Fig. 13, Fig. 13(a) represents the variation of output per unit of installed capacity of wind and PV power generation equipment with time during a typical day in North China, carved through the methods mentioned in Ref. [42]. and Fig. 13(b) demonstrates the total amount of renewable energy that can be consumed by DC under different scenarios, and the enhancement of renewable energy consumption ability by the utilization of flexible resources, and the bar charts from left to right for the same scenario indicate the scenarios of PV only, both PV and WP, and WP only, respectively. The use of electricity flexibility can shift the renewable power in time to meet the DC's own load condition through the power storage equipment; the use of heat flexibility can adjust the DC's load condition by regulating the energy consumption of the cooling system to meet the renewable power; the use of computing flexibility can shift and adjust the load of DC at the same time to achieve the maximization of renewable energy consumption. The utilization of the three types of flexibility resources can enhance the renewable energy consumption ability of DC by 13.7 %, 8.5 %, and 15.3 %, respectively. And the utilization of all the three types of flexibility resources can enhance the renewable energy consumption ability by up to 19.5 %, compared with S1, which does not utilize the DC's own flexibility resources at all.

For S8, which utilizes three types of flexibility resources at the same time, the impact of renewable power on its carbon emission, energy consumption and economic cost is shown in Fig. 14, with the white area representing that the installed capacity of renewable power is too large for the DC to consume. It can be seen that with the increase of wind power installed capacity, although the energy consumption of the DC will increase to consume more renewable power, up to 37.8 %, but overall, its carbon emission and economic cost have a significant decrease, the economic cost of up to 63.9 % lower, and to achieve a net-zero emission. The installed cost on the economic cost side obviously increases as installed capacity increases, but the reduction in the cost of energy supply effectively makes up for this.

### 4.4. Sensitivity analysis of heat flexibility

In the original scenario, each server rack is configured with 17 servers, with the rack occupancy rate set at 100 %. For the same number of servers, a lower rack occupancy rate indicates that more racks are required, which increases the DC's floor area, virtual energy storage capacity, and reduces the pressure on the cooling system. To explore the impact of this factor on the utilization of flexibility resources in DCs, this paper calculates scenarios with different rack occupancy rates and total rack quantities. The calculation results are shown in Fig. 15(a), and Fig. 15(b) illustrates the temperature changes in the DC for some occupancy rate scenarios.

The results show that for the same number of servers, DCs with lower rack occupancy rates do indeed have higher virtual energy storage capacity and greater heat flexibility. However, when considering the cost of the racks themselves and the construction costs, the benefits brought by higher heat flexibility are far outweighed by the additional costs. Therefore, within the power capacity allowed by the cooling system, DCs with higher density and higher rack occupancy rates are more advantageous economically.

## 5. Conclusion

In summary, this paper provides a detailed description of the computing-heat coupling mechanism and establishes a comprehensive flexibility resource planning and scheduling strategy for independent DCs. Based on the optimization scheduling results, the paper reveals the coordinated interaction between the three types of flexibility resources. The main contributions of the study are as follows:

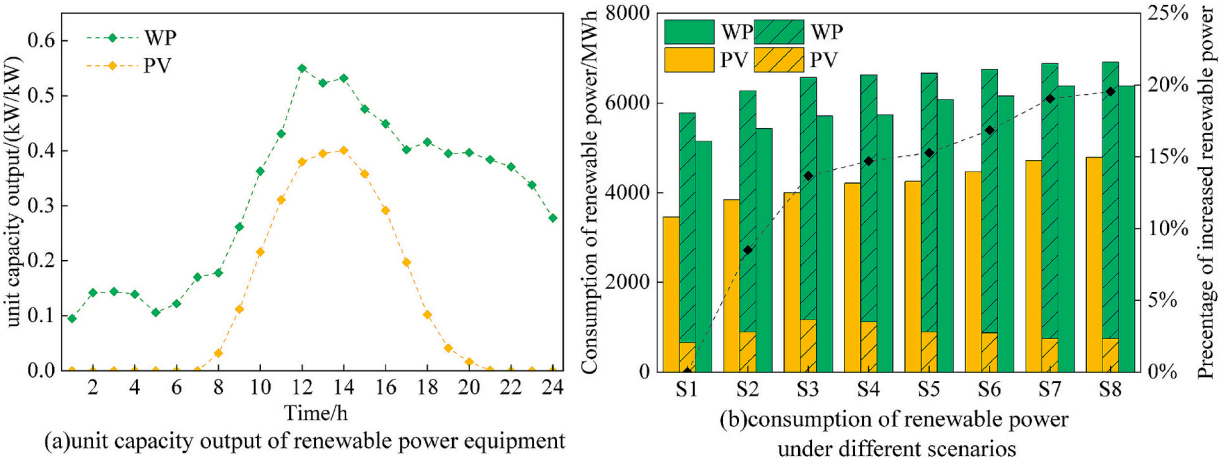
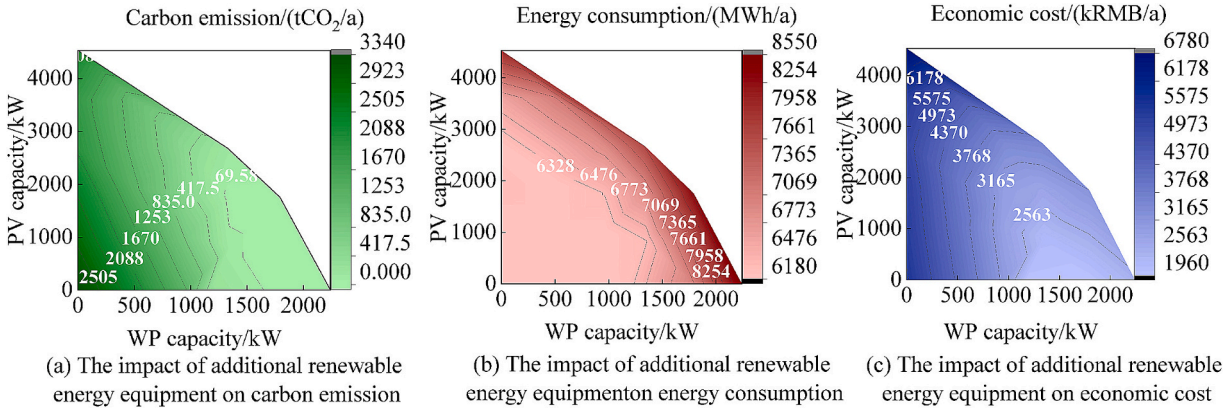



**Fig. 13.** Wind and solar output curves and renewable energy consumption ability of DC.

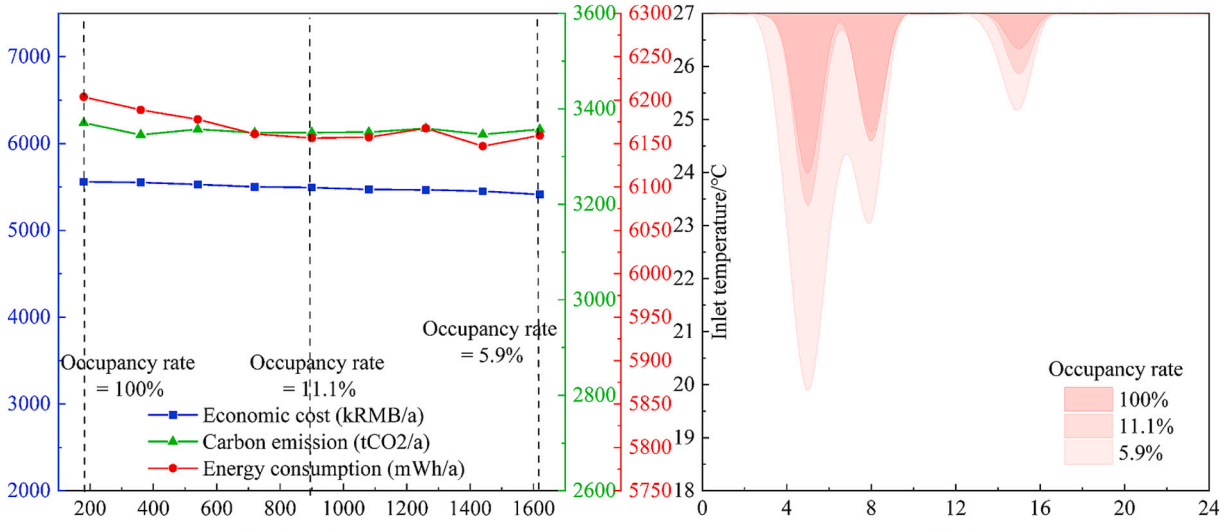



(a) The impact of additional renewable energy equipment on carbon emission

(b) The impact of additional renewable energy equipmenton energy consumption

(c) The impact of additional renewable energy equipment on economic cost

**Fig. 14.** The impact of additional renewable energy equipment on DC operation.




(a)The economic cost, carbon emission and energy consumption changes by the number of racks

(b)The inlet temperature changes for some occupancy rate scenarios

**Fig. 15.** The impact of number of racks and occupancy rate on the utilization of flexibility resources.

1) This paper analyzes the computing-heat coupling mechanism in depth and clarifies the significant impact of temperature and load rate on the overall energy consumption and computing efficiency of DCs. The research results indicate that whether or not the coupling mechanism between computing and cooling systems is considered will have a significant impact on the planning and scheduling of DCs.
2) A comprehensive flexibility resource planning and scheduling strategy for DCs is established. Simulation results show that compared to cases where no flexibility resources are utilized, this strategy can reduce carbon emissions by 38.5 %, energy consumption by 22.8 %, and economic costs by 21.6 %, while improving DC's PUE and computing efficiency indicators. At the same time, utilizing multiple types of flexibility resources can effectively enhance energy-related metrics in DCs. However, due to conflicts in the utilization mechanisms of multiple flexibility resources—such as increasing the air supply temperature, which reduces cooling system energy consumption but increases server energy consumption—there are certain limitations in the synergistic use of multiple flexibility resources.
3) The DC's comprehensive flexibility resource utilization strategy is investigated under different scenarios of electricity prices, workload states, and rack occupancy rates using sensitivity analysis. The study reveals the impact of the workload type distribution and the maximum delay time on the energy consumption of the DC. Specifically, when the average grid price is higher than 0.72 RMB/kWh, utilizing gas turbines for constant power supply can have a positive effect on DC operation, while also stimulating the potential for time transfer in computing-side flexibility resources. The average maximum delay time and proportion of batch workload have a significant impact on DC energy consumption. When the batch workload proportion is 50 %, and the average maximum latency exceeds 7.5 h, the energy consumption of the DC is minimized. Additionally, an optimal matching relationship exists between the batch workload proportion and the maximum delay time. When the average maximum latency exceeds 12.5 h, and the proportion of interactive workload is 70 %, the overall energy consumption of the DC decreases from 6130 MWh per year to 5816 MWh per year, reducing energy consumption by an additional 6 %. The sensitivity analysis on the heat side indicates that high-density and high-rack-occupancy DCs should have a greater economic advantage.

Meanwhile, this study also presents the following limitations that warrant further investigation:

1) The current framework primarily focuses on countermeasures for DCs to address uncertain loads, such as reserved power generation capacity and battery storage configurations. However, it lacks systematic characterization of uncertainties inherent in computing workload and renewable energy integration. Future research will undertake comprehensive uncertainty quantification and incorporate robust optimization methods to enhance decision-making under stochastic conditions.
2) The flexibility resources considered are limited to conventional types typically deployed in single DC operations. Emerging flexibility mechanisms, including waste heat recovery systems and cross-regional workload migration between geographically distributed DCs, remain unexplored. Subsequent studies will expand the scope of flexibility resources orchestration through multidimensional optimization frameworks that account for both temporal and spatial coordination potentials.

## CRediT authorship contribution statement

**Yongzhen Wang:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Jiayu Lin:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Yibo Han:** Writing – review & editing, Software, Formal analysis. **Kai Han:** Supervision, Resources, Funding acquisition. **Juntao Han:** Writing – review & editing, Software. **Te Han:** Writing – review & editing. **Yiming Wei:** Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A

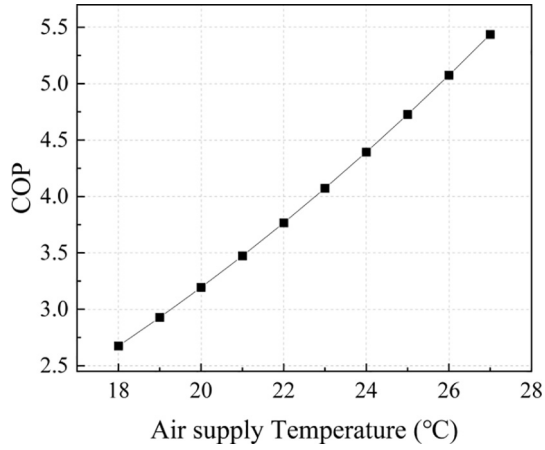



**Fig. A1.** COP under different air supply temperature.

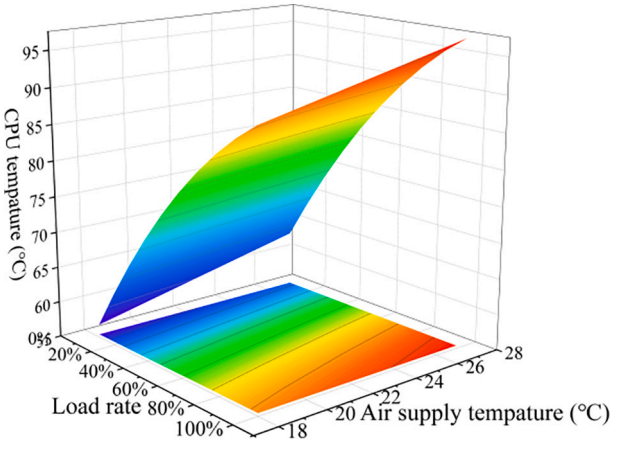



**Fig. A2.** CPU temperature under different load rates and air supply temperature.

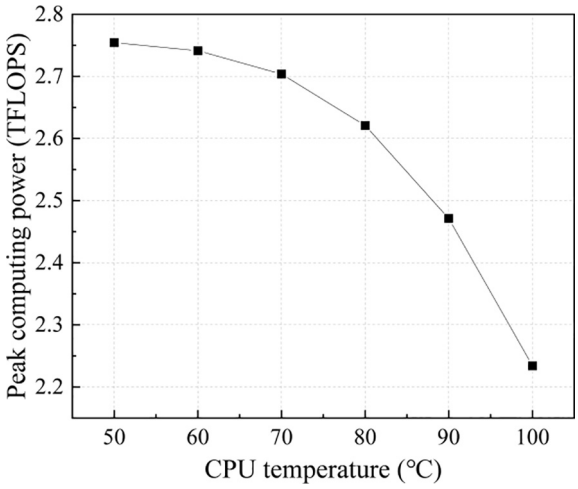



**Fig. A3.** Peak computing power under different CPU temperature.

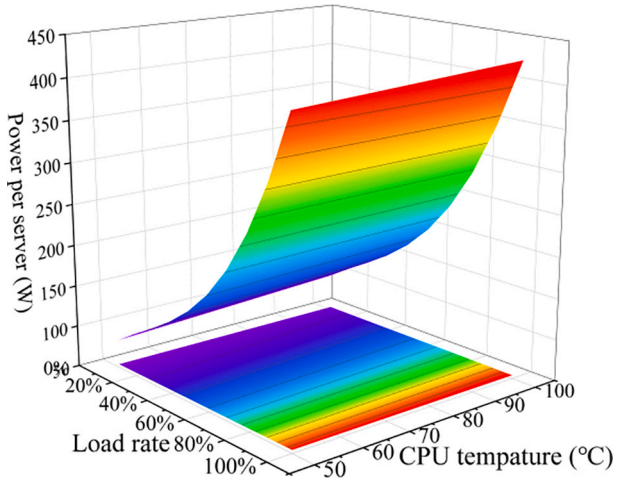



**Fig. A4.** Power of servers under different load rates and CPU temperature.

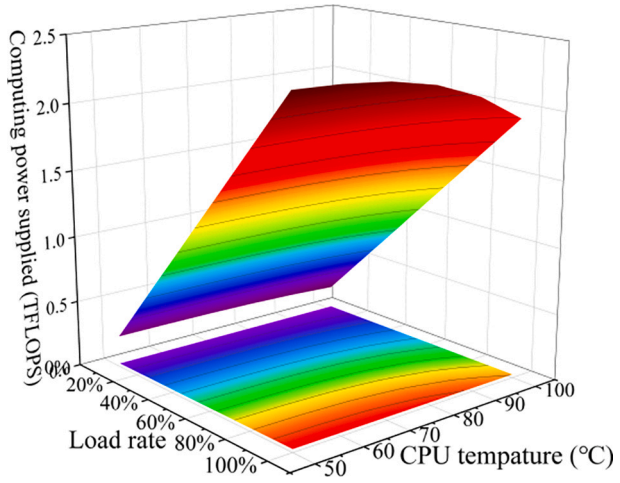



**Fig. A5.** Computing power supplied under different load rates and CPU temperature.

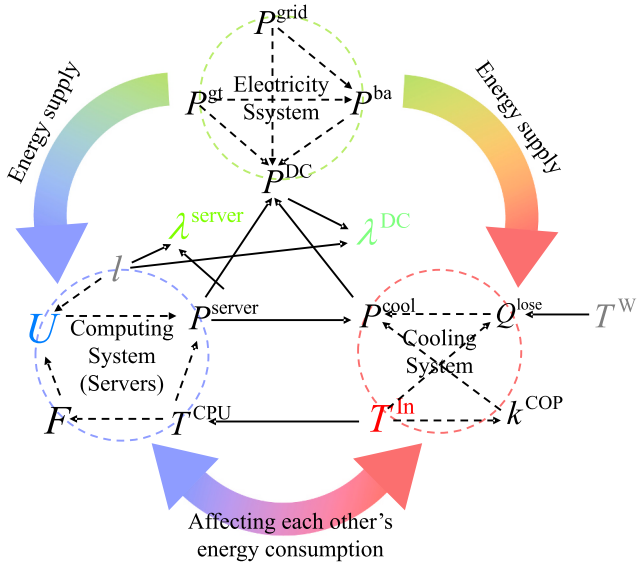



**Fig. A6.** Flexibility resources collaborative framework.

## Appendix B

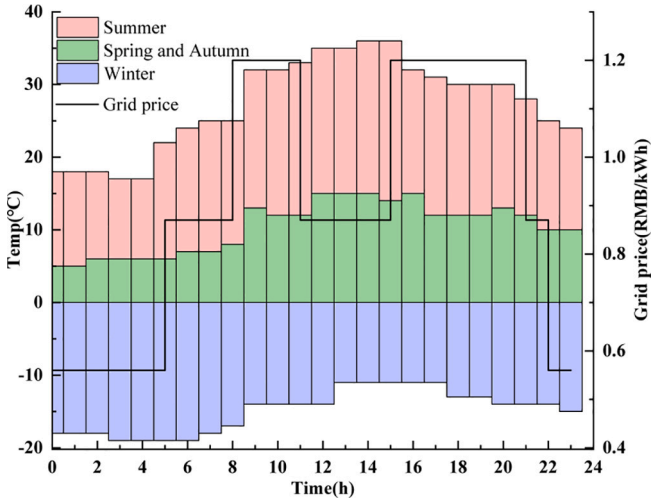
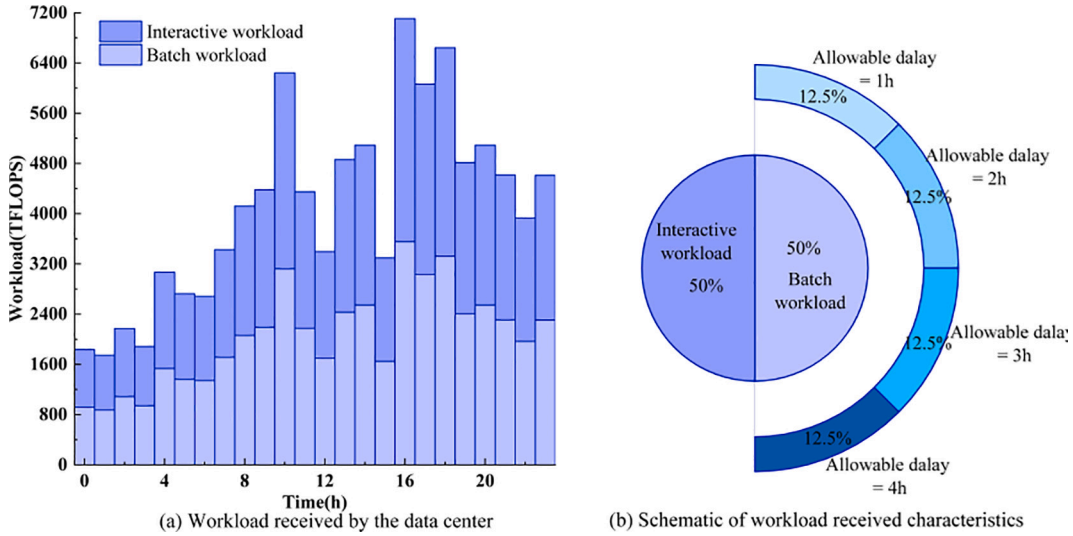



**Fig. B1.** The external temperature and grid price in typical days in different seasons.




(a) Workload received by the data center (b) Schematic of workload received characteristics

**Fig. B2.** The workload received by the DC.

**Table B1**
The parameters of the server.

| Parameters | Value |
|---|---|
| CPU | Intel Xeon Platinum 8280 28-Core, 2.7GHz, 38.5 MB L3 Cache |
| Peak computational power | 2.4TFLOPS@FP32 |
| Ideal power | 51.7 W |
| Peak power | 421 W |
| Number | 3060 |
| Run year | 10年 |

**Table B2**
The coefficients of the DC.

| Fitting coefficients | Value | Coefficients | Value | Unit |
|---|---|---|---|---|
| $b_1$ | −3.07 × 10–6 | r | 0.06 | – |
| $b_2$ | 4.47 × 10–4 | η | 1.6 | – |
| $b_3$ | −2.24 × 10–2 | $\chi_1$ | −0.2 | – |
| $b_4$ | 2.78 | $\chi_2$ | 0 | – |
| $c_1$ | 421 | $\chi_3$ | 0.02 | – |
| $c_2$ | −314 | α | 0.816 | |
| $c_3$ | 244 | k | 0.5 | – |
| $c_4$ | 47.627 | $\theta_1$ | 0.8 | $W/(m^2K)$ |
| $c_5$ | 68.924 | $\theta_2$ | 0.8 | $W/(m^2K)$ |
| $k_1$ | 0.0068 | S | 1000 | $m^2$ |
| $k_2$ | 0.0008 | $C^w$ | 1.005 | kJ/(kgK) |
| $k_3$ | 0.458 | $\rho^w$ | 1.1748 | $kg/m^3$ |
| $a_1$ | 75 | $V^w$ | 1250 | $m^3$ |
| $a_2$ | 3 | $C^{DC}$ | 1.005 | kJ/(kgK) |
| $g_1$ | 0.41 | $\rho^{DC}$ | 1.1748 | $kg/m^3$ |
| $g_2$ | −0.94 | $V^{DC}$ | 1000 | $m^3$ |
| $g_3$ | 0.38 | $\sum_j (d_j/\sigma_j)$ | 14.71 | $m^2K/W$ |
| $g_4$ | 0.019 | | | |

**Table B3**
The parameters of the energy equipment.

| Device | Efficiency/COP | Investment cost(RMB/kW) |
|---|---|---|
| WP | – | 9400 |
| PV | – | 7800 |
| BA | – | 1800 |
| GT | $0.41x_{gt}^3 - 0.94x_{gt}^2 + 0.83x_{gt} + 0.019$ | 1100 |
| Cooling system | $0.0068\left(T_t^{in}\right)^2 + 0.0008T_t^{in} + 0.458$ | 600 |

## Appendix C

**Table C1**
The optimization results of all cases.

| Optimization objective | Parameter | S1 | S2 | S3 | S4 | S5 | S6 | S7 | S8 |
|---|---|---|---|---|---|---|---|---|---|
| Energy consumption | Energy consumption ($\times10^3$MWh/a) | 7.93 | 6.89 | 7.94 | 6.89 | 7.04 | 6.13 | 7.04 | 6.13 |
| | Carbon emission ($\times10^3$tCO$_2$/a) | 4.52 | 3.93 | 3.99 | 3.49 | 4.03 | 3.50 | 3.58 | 3.12 |
| | Economic cost ($\times10^6$RMB/a) | 7.77 | 6.77 | 8.00 | 6.97 | 6.77 | 5.91 | 7.02 | 6.10 |
| | PUE | 1.37 | 1.18 | 1.37 | 1.18 | 1.37 | 1.18 | 1.37 | 1.18 |
| | Computing efficiency ($\times10^{-3}$TFLOPS/W) | 5.65 | 6.50 | 5.64 | 6.50 | 6.36 | 7.30 | 6.36 | 7.30 |
| Carbon emission | Energy consumption | 7.93 | 6.89 | 7.95 | 6.90 | 7.04 | 6.13 | 7.09 | 6.13 |
| | Carbon emission | 4.52 | 3.93 | 3.55 | 3.10 | 4.02 | 3.49 | 3.17 | 2.74 |
| | Economic cost | 7.77 | 6.77 | 9.08 | 7.95 | 6.78 | 5.92 | 8.10 | 7.02 |
| | PUE | 1.37 | 1.18 | 1.37 | 1.18 | 1.37 | 1.18 | 1.37 | 1,18 |
| | Computing efficiency | 5.65 | 6.50 | 5.63 | 6.49 | 6.36 | 7.30 | 6.31 | 7.30 |
| Economic cost | Energy consumption | 7.93 | 6.89 | 7.94 | 6.89 | 7.11 | 6.18 | 7.11 | 6.20 |
| | Carbon emission | 4.52 | 3.93 | 4.21 | 3.70 | 4.06 | 3.52 | 3.81 | 3.33 |
| | Economic cost | 7.77 | 6.77 | 7.45 | 6.46 | 6.75 | 5.89 | 6.46 | 5.63 |
| | PUE | 1.37 | 1.18 | 1.37 | 1.18 | 1.37 | 1.18 | 1.37 | 1.18 |
| | Computing efficiency | 5.65 | 6.50 | 5.64 | 6.50 | 6.30 | 7.24 | 6.30 | 7.22 |

**Table C2**
The capacity scheduling results of all cases.

| Optimization objective | Capacity(kW) | S1 | S2 | S3 | S4 | S5 | S6 | S7 | S8 |
|---|---|---|---|---|---|---|---|---|---|
| Energy consumption | Gas turbine | 940 | 940 | 1470 | 1300 | 770 | 770 | 1200 | 940 |
| | Battery | 240 | 240 | 1320 | 1320 | 200 | 200 | 1170 | 1170 |
| | Cooling system | 940 | 940 | 940 | 940 | 770 | 770 | 760 | 770 |
| Carbon emission | Gas turbine | 940 | 940 | 1020 | 930 | 770 | 780 | 850 | 790 |
| | Battery | 240 | 240 | 1270 | 1190 | 200 | 200 | 1160 | 1000 |
| | Cooling system | 940 | 940 | 940 | 930 | 770 | 780 | 770 | 790 |
| Economic cost | Gas turbine | 940 | 940 | 1430 | 1050 | 810 | 770 | 1080 | 930 |
| | Battery | 240 | 240 | 1320 | 1320 | 210 | 200 | 1180 | 1190 |
| | Cooling system | 940 | 940 | 930 | 930 | 810 | 770 | 810 | 800 |

## Data availability

Data will be made available on request.

## References


[1] Mytton D, Ashtine M. Sources of data center energy estimates: a comprehensive review. Joule 2022;6(9):2032–56.
[2] Wang Y, Han Y, Han K, et al. Does DeepSeek curb the surge of energy consumption in data centers? The. Innovation 2025:100944.
[3] Agency I E.Empowering variable renewables-options for flexible electricity systems. OECD Publishing; 2009. p. 11–3.
[4] Cao Y, Cheng M, Zhang S, et al. Data-driven flexibility assessment for internet data center towards periodic batch workloads. Appl Energy 2022;324:119665.
[5] Jian J, Zhao J, Ji H, et al. Supply restoration of data centers in flexible distribution networks with spatial-temporal regulation. IEEE Transactions on Smart Grid 2024; 15(1):340–54.
[6] Wu S, Wang Q, Chen Q, et al. Cyber-physical integrated planning of distribution networks considering spatial-temporal flexible resources. Protection And Control of Modern Power Systems 2024;9(3):142–56.

[7] Chen D, Ma Y, Wang L, et al. Spatio-temporal management of renewable energy consumption, carbon emissions, and cost in data centers. Sustainable Computing: Informatics and Systems 2024;41:100950.
[8] Zhou S, Zhou M, Wu Z, et al. Dual-drive coordinated operation strategy for internet data centers by exploiting spatiotemporal flexibility. J Clean Prod 2024;450:141741.
[9] Zhou S, Zhou M, Wu Z, et al. Energy-aware coordinated operation strategy of geographically distributed data centers. Int J Electr Power Energy Syst 2024;159:110032.
[10] Dvorkin V. Agent coordination via contextual regression (AgentCONCUR) for data center flexibility. IEEE Trans Power Syst 2024:1–11.
[11] Jin C, Bai X, Yang C, et al. A review of power consumption models of servers in data centers. Appl Energy 2020;265:114806.
[12] Basmadjian R, Ali N, Niedermeier F, et al. A methodology to predict the power consumption of servers in data centres. ACM Sigcomm International Conference on Energy-efficient Computing & Networking 2011. https://doi.org/10.1145/2318716.2318718.
[13] Arroba P, Risco-Martin JL, Zapater M, et al. Server power modeling for run-time energy optimization of cloud computing facilities62. 6th International Conference on Sustainability in Energy and Buildings; 2014. p. 401–10.
[14] Kavanagh R, Djemame K. Rapid and accurate energy models through calibration with IPMI and RAPL. Concurrency and Computation: Practice and Experience 2019;31(13):1–21.
[15] Rezaei-Mayahi M, Rezazad M, Sarbazi-Azad H. Temperature-aware power consumption modeling in Hyperscale cloud data centers. Future Generation Computer Systems-the International Journal of Escience 2019;94:130–9.
[16] Fan X, Weber W, Barroso L. Power provisioning for a warehouse-sized computer. Isca'07: 34th Annual International Symposium on Computer Architecture, Conference Proceedings; 2007. p. 13–23.
[17] Zhang X, Lu J, Qin X, et al. A high-level energy consumption model for heterogeneous data centers[J]. Simul Model Pract Theory 2013;39(1):41–55.
[18] SPEC. SPEC CPU® 2006 Results – Query. https://www.spec.org/cgi-bin/osgresults; 2006. Accessed January 14,2025.
[19] Lin W, Wang W, Wu W, et al. A heuristic task scheduling algorithm based on server power efficiency model in cloud environments. Sustainable Computing: Informatics and Systems 2018;20:56–65.
[20] Vasques TL, Moura P, Almeida A. A review on energy efficiency and demand response with focus on small and medium data centers. Energ Effic 2019;12(5):1399–428.
[21] Cui Y, Cheng Y, Zhu H, et al. Stochastic optimization for capacity configuration of data center microgrid thermal energy management equipment considering flexible resources. Int J Electr Power Energy Syst 2024;160:110132.
[22] Ji H, Chen S, Hao Yu, et al. Robust operation for minimizing power consumption of data centers with flexible substation integration. Energy 2022;248:123599.
[23] Van Damme T, De Persis C, Tesi P. Optimized thermal-aware job scheduling and control of data centers. IEEE Trans Control Syst Technol 2019;27(2):760–71.
[24] Wang T, Yu H, Zhao J, et al. Optimal energy management of data centers considering the synergy of computing power and thermal power flexibility. Gaodianya Jishu/High Voltage Eng 2024;50(9):4069–79.
[25] Li W, Qian T, Zhao W, et al. Decentralized optimization for integrated electricity–heat systems with data center based energy hub considering communication packet loss. Appl Energy 2023;350:121586.
[26] Li W, Qian T, Zhang Y, et al. Distributionally robust chance-constrained planning for regional integrated electricity–heat systems with data centers considering wind power uncertainty. Appl Energy 2023;336:120787.
[27] Ham SW, Kim MH, Choi BN, et al. Simplified server model to simulate data center cooling energy consumption. Energy Build 2015;86(1):328–39.
[28] Zhu J, Liu D, Li B, et al. Data center server power consumption model based on temperature estimation considering CPU working state. Dianli Xitong Zidonghua/Automation of Electric Power Systems 2023;47(14):140–8.
[29] S Pagani, P. D. S Manoj, A Jantsch, et al. Machine learning for power, energy, and thermal management on multicore processors: a survey. IEEETrans Comput-Aided Des Integr Circuits Syst 2020; 39: 101–116.
[30] Guo CR, Yu YT, Luo J, et al. Effects of temperature and utilization on CPU performance (in Chinese). Microelectronics 2018;48:542–7.
[31] Hu M, He Y, Lin X, et al. Digital twin model of gas turbine and its application in warning of performance fault. Chin J Aeronaut 2023;36(3):449–70.
[32] Taghizad-Tavana K, Tarafdar-Hagh M, Nojavan S, et al. Operation of smart distribution networks by considering the spatial-temporal flexibility of data centers and battery energy storage systems. Sustain Cities Soc 2024;114:105746.
[33] Wang H, Zhu L, Sun W, et al. Research on energy pricing of the hydrogen refueling station based on master-slave game in multi-market. Appl Energy 2024;373:123906.
[34] ASHRAE. Thermal guidelines for data processing environments. 5th ed. Georgia, USA: ASHRAE; 2021. p. 10–40.
[35] Han J, Yan Y, Wang Y, et al. Dual-time scale collaborative optimization of data center energy system: considering multi-task response mechanism and hybrid hydrogen-battery energy storage. J Energy Storage 2025;119:116244.
[36] Wei L, Ai X, Fang J, et al. Data-augmentation acceleration framework by graph neural network for near-optimal unit commitment. Appl Energy 2025;377:124332.
[37] Wang H, Zhu L, Sun W, et al. Research on energy pricing of the hydrogen refueling station based on master-slave game in multi-market. Appl Energy 2024;373:123906.
[38] Han J, Han K, Wang Y, et al. Multi-objective planning and sustainability assessment for integrated energy systems combining ORC and multi-energy storage: 4E (economic, environmental, exergy and emergy) analysis.Case studies. Therm Eng 2025;65:105674.
[39] Radovanović A, Koningstein R, Schneider I, et al. Carbon-aware computing for datacenters. IEEE Trans Power Syst 2022;38(2):1270–80.
[40] Han J, Han K, Han T, et al. Data-driven distributionally robust optimization of low-carbon data center energy systems considering multi-task response and renewable energy uncertainty. J Build Eng 2025;102:111937.
[41] Sun T, Huang X, Liang C, et al. Energy consumption and energy saving analysis of air-conditioning Systems of Data Centers in typical cities in China. Sustainability 2023;15(10):7826.
[42] Duarte JLR, Fan N. Operations of data centers with onsite renewables considering greenhouse gas emissions. Sustainable Computing: Informatics and Systems 2023;40:100903.